Continuous coupling trajectories — run report

RUN

191137_cross-sweep_quick-test-phi2
Status: completed | Started: 2026-09-29T18:11:37.347928+00:00

DIAGNOSTIC SUMMARY

Target actuator(s): phi2; sine: 0.1164 Hz; measured PAX rate: 6.86 Hz; saved RP reference rate: 381 Hz; estimated PAX samples/cycle: 58.96.
Recorded cycles: 3; bias values: phi1 target 0, phi2 target 11. Waveform tracking: deviations or missing/clipped reference present. Rising/falling branches (>5° descriptive threshold): strong separation present. Cycle repeatability (5° descriptive threshold): nonrepeatable above threshold. Bias-dependent plane changes (>5° descriptive threshold): obvious changes present. Low-DoP intervals: True. These thresholds are flags, not mechanism assignments.

PURPOSE

Continuously sine-drive phi1, phi2, or both while stepping and settling the orthogonal bias; record PAX plus measured OUT1/OUT2 references.

REQUIRED OPTICAL SETUP

PAX at final output; both paths open unless prompted. Wire OUT1 to IN1 and OUT2 to IN2 as voltage references. 'both' runs the complete diagnostic for both actuators automatically.

RECORDED BENCH CONTEXT (RECIPE)

In front of NPBS 1, C port
Red Pitaya (0-1 V command) -> Thorlabs MDT690 -> piezo

RUN COMMENT

dry run 2 except with phi2 now

OPERATOR SETUP NOTES

V_pi expectation only; verify channel and measured driver gain before calibration.

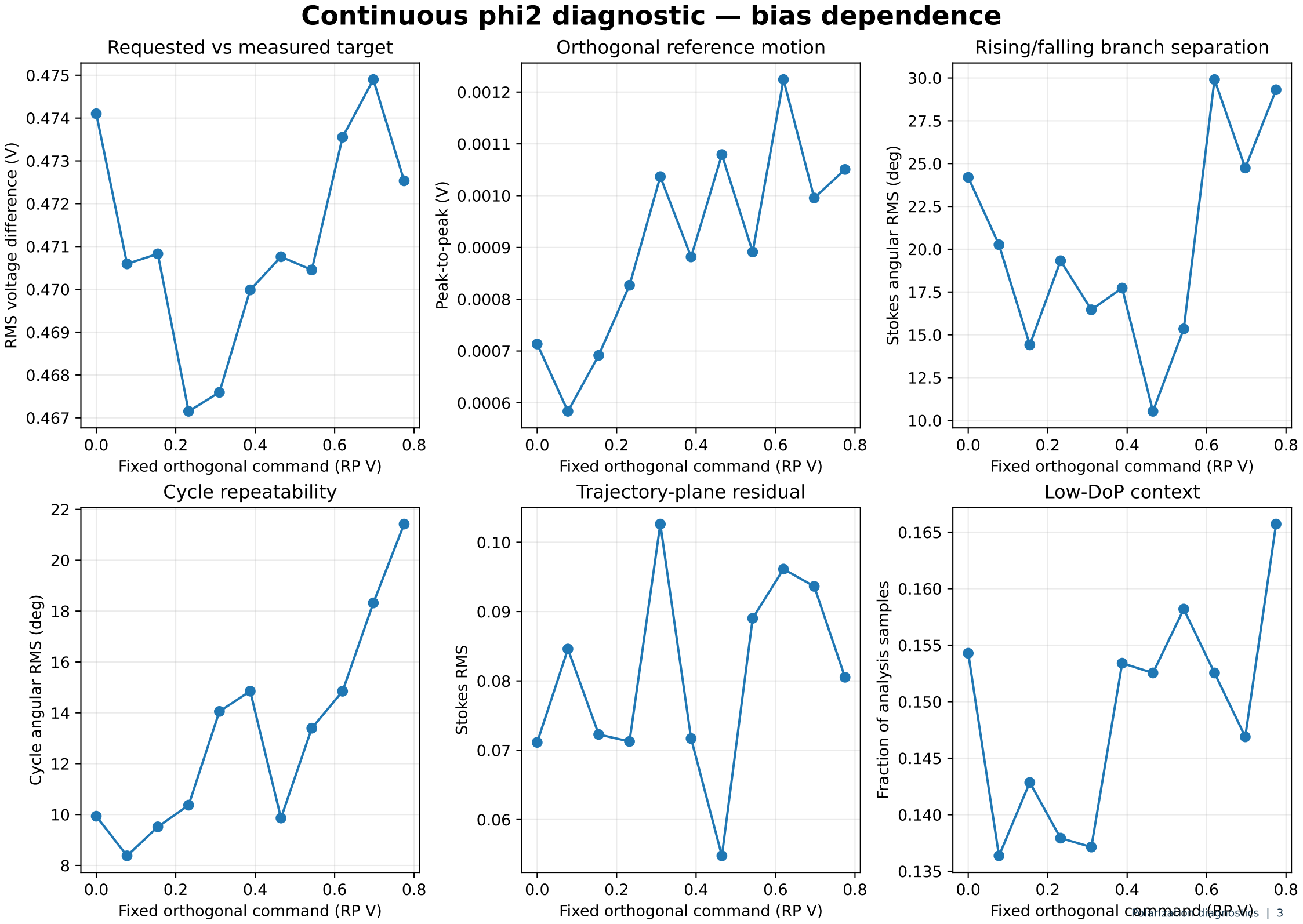
DATA

Continuous coupling trajectories — run report

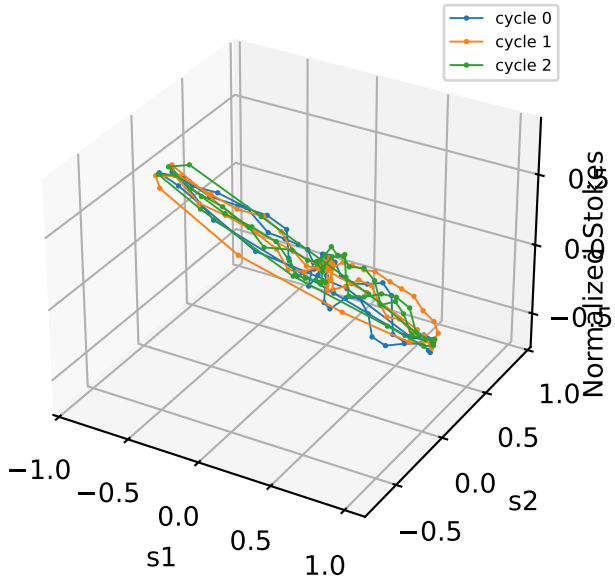
2,679 recorded samples; acquisition: 2,679
Source: data.csv; full acquisition settings: recipe.json; status: run.json

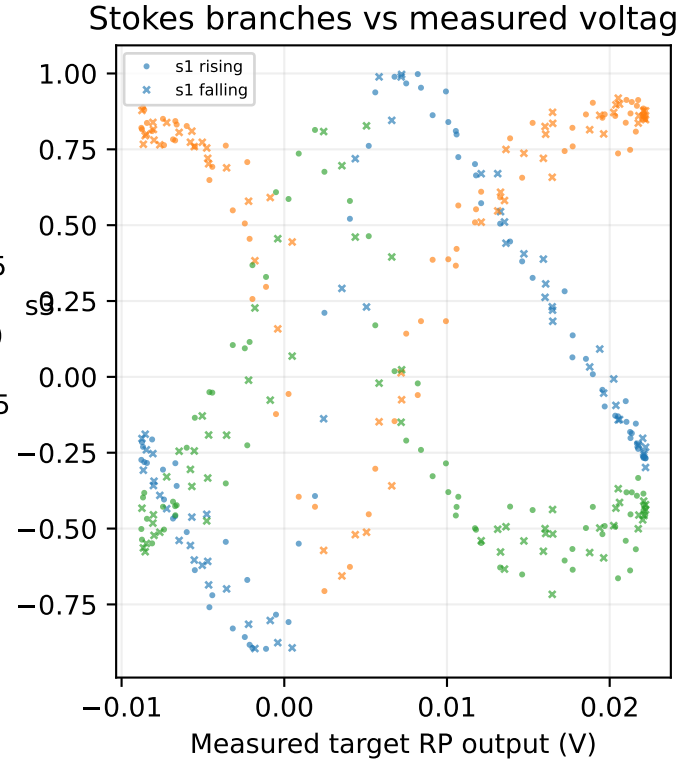
RAW DOP

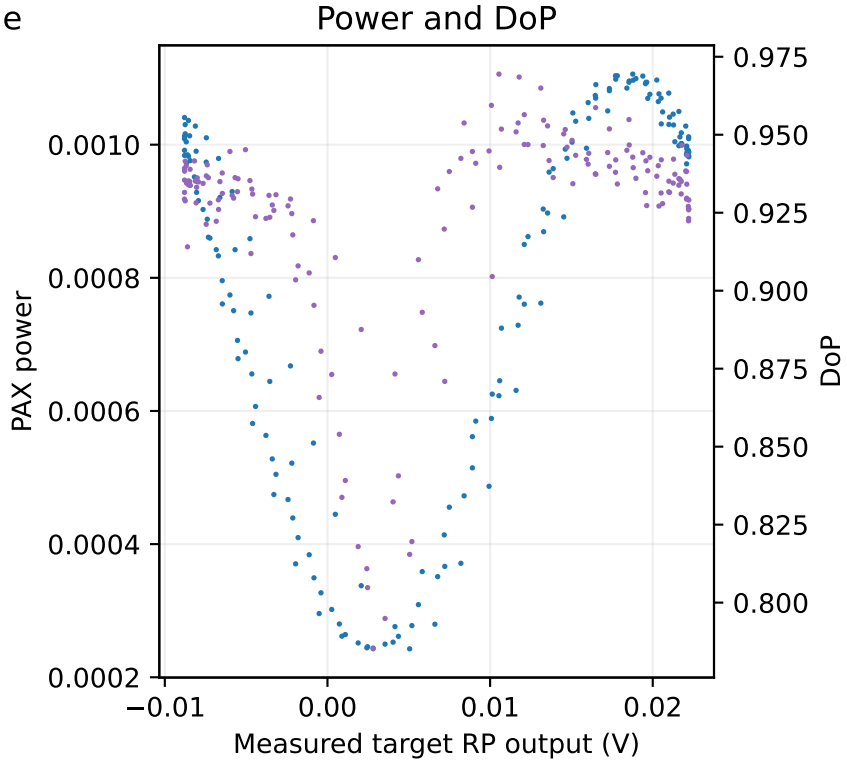
Mean 0.9254; range 0.7516 to 3.3933. 6 samples outside 0–1. Telemetry is not a new calibration.

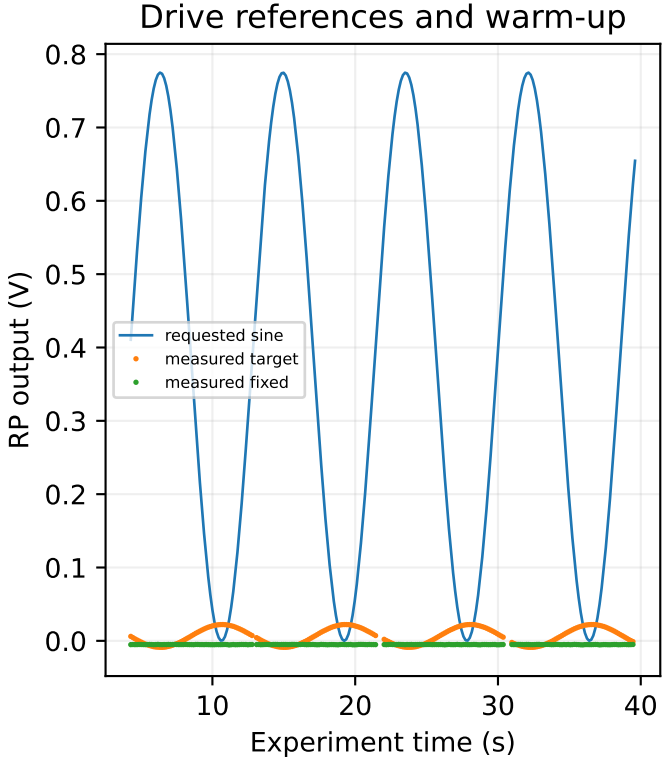

Continuous phi2 diagnostic — bias dependence
Requested vs measured target
RMS voltage difference (V)
Fixed orthogonal command (RP V)
Orthogonal reference motion
Peak-to-peak (V)
Fixed orthogonal command (RP V)
Rising/falling branch separation
Stokes angular RMS (deg)
Fixed orthogonal command (RP V)
Cycle repeatability
Cycle angular RMS (deg)
Fixed orthogonal command (RP V)
Trajectory-plane residual
Stokes RMS
Fixed orthogonal command (RP V)
Low-DoP context
Fraction of analysis samples
Fixed orthogonal command (RP V)

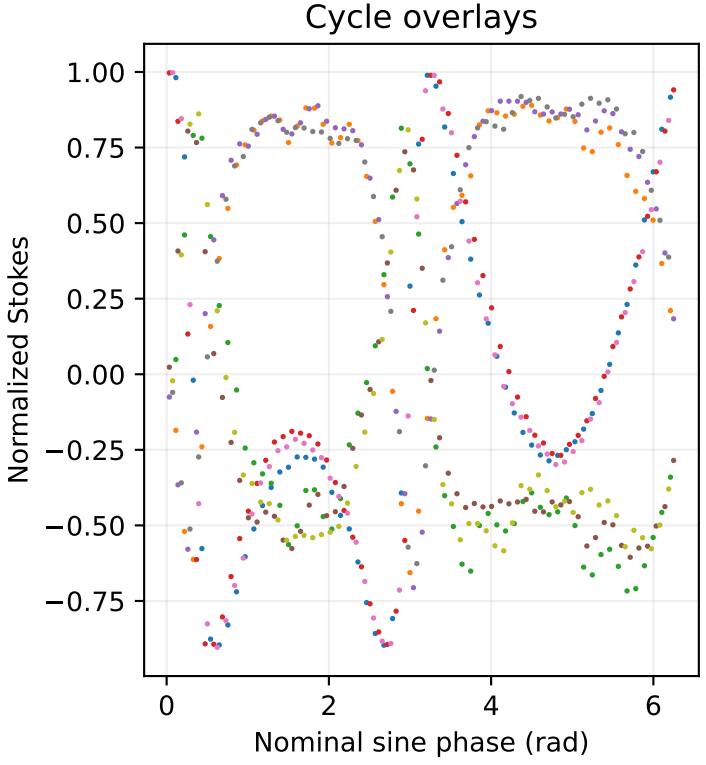
phi2 sine; fixed phi1 = 0 RP V (segment 0)





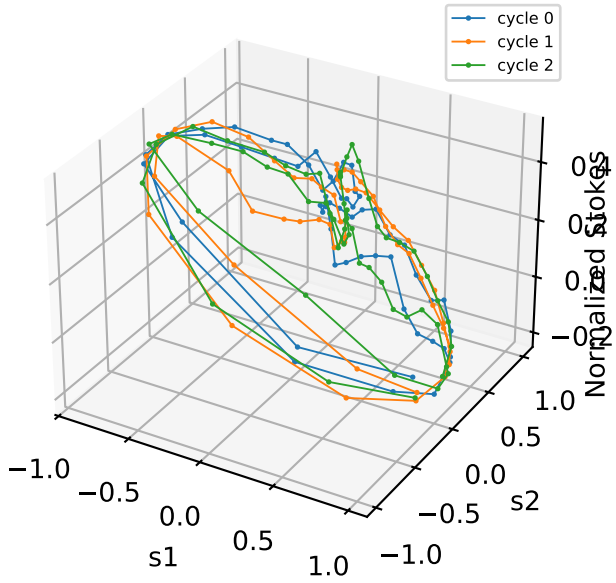


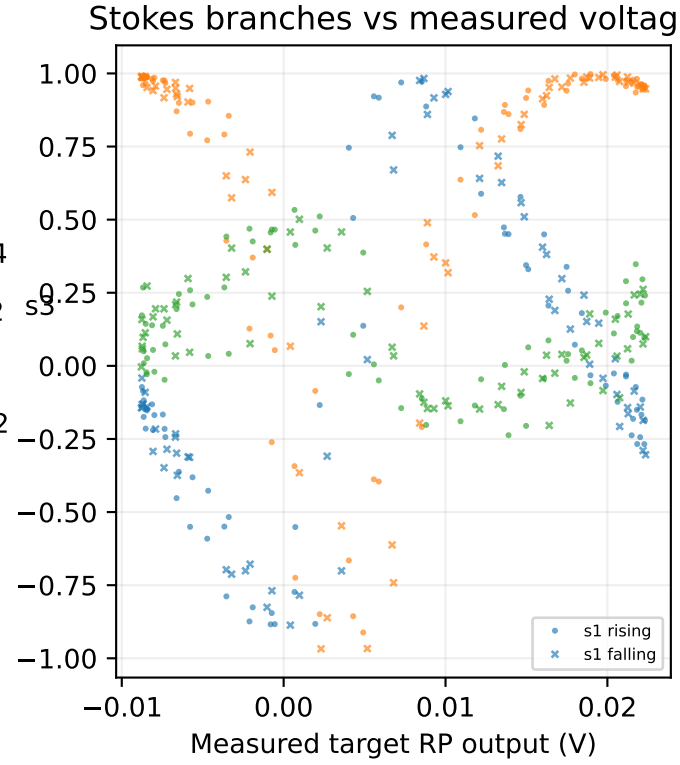


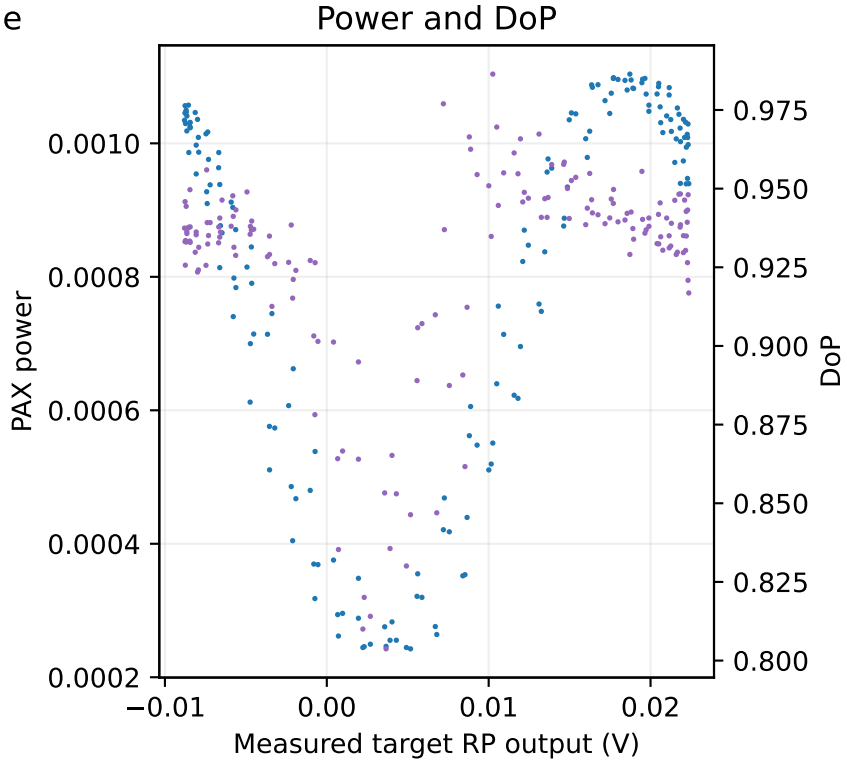


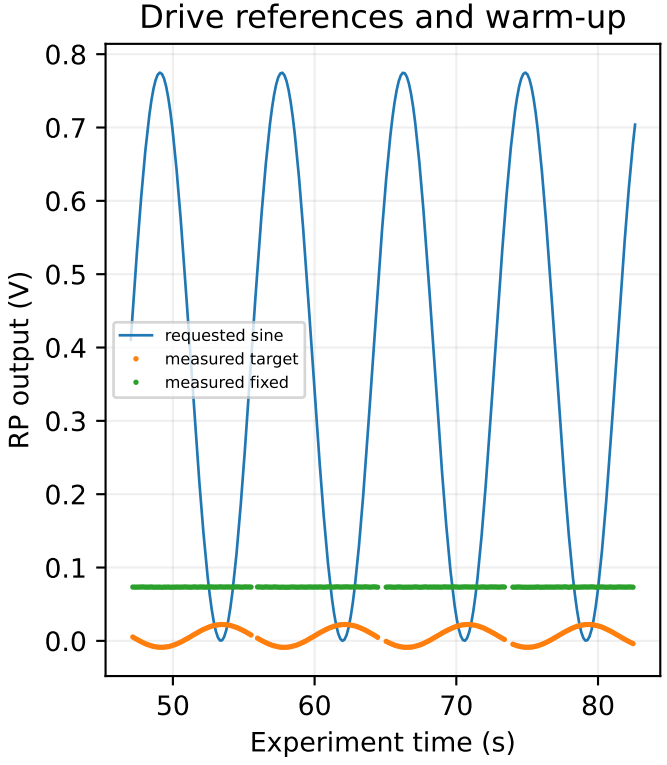
Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.254, 0.667, 0.701]
Plane residual RMS: 0.07114
Rising/falling angular RMS: 24.2°
Rising/falling power RMS: 7.47e-05 PAX units
Cycle angular RMS: 9.94°
Cycle power RMS: 4.73e-05 PAX units
Cycle DoP RMS: 0.0125
Tracking RMS: 0.4741 V
Reference coverage: 96.0%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.29, 0.67, 0.68], residual 0.069
Cycle 1 plane: [0.23, 0.65, 0.72], residual 0.0666
Cycle 2 plane: [0.24, 0.67, 0.70], residual 0.0672

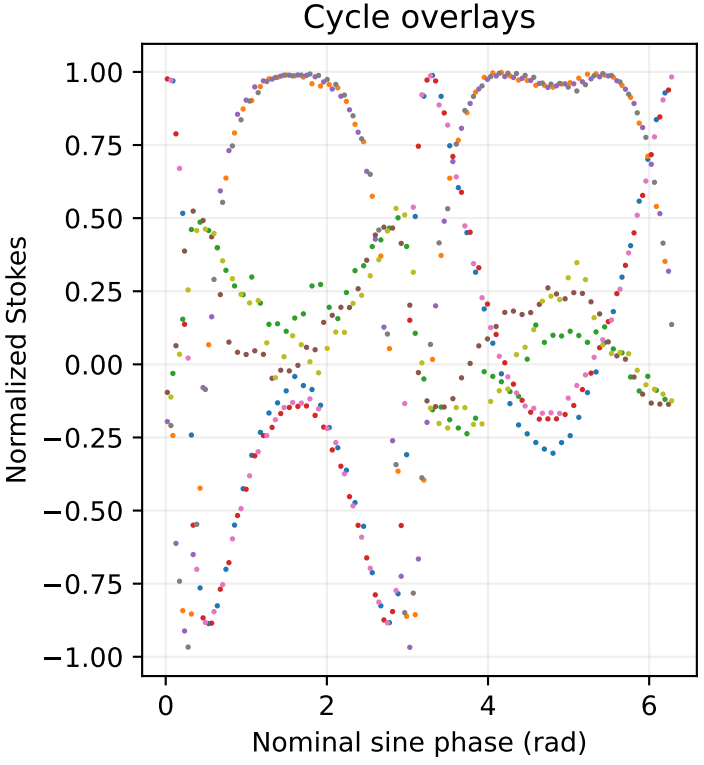
phi2 sine; fixed phi1 = 0.07739 RP V (segment 1)





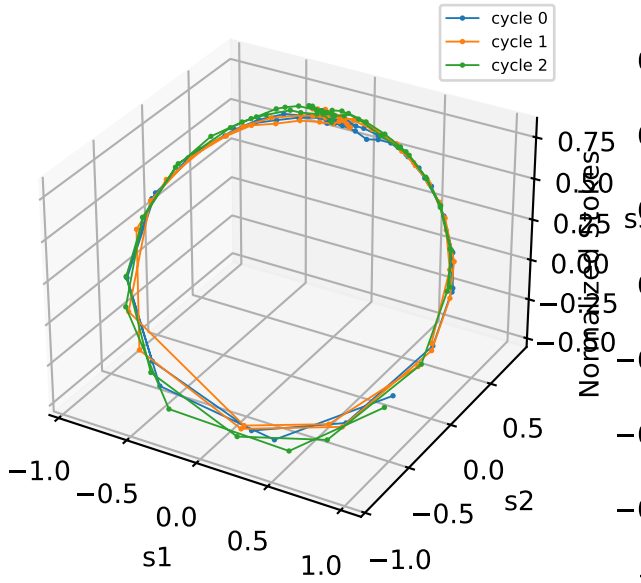




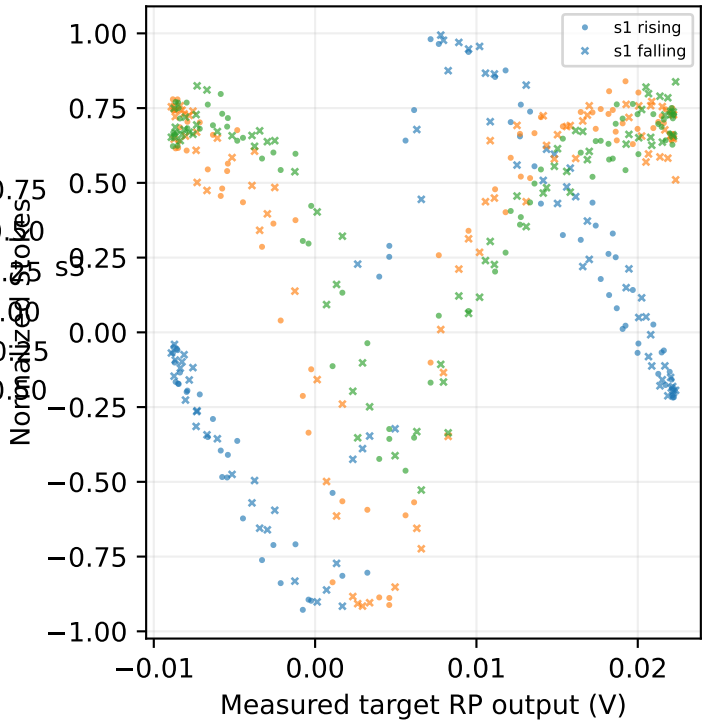


Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.301, 0.137, 0.944]
Plane residual RMS: 0.0846
Rising/falling angular RMS: 20.3°
Rising/falling power RMS: 8.35e-05 PAX units
Cycle angular RMS: 8.38°
Cycle power RMS: 2.88e-05 PAX units
Cycle DoP RMS: 0.0144
Tracking RMS: 0.4706 V
Reference coverage: 95.5%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.32, 0.14, 0.94], residual 0.0755
Cycle 1 plane: [0.28, 0.12, 0.95], residual 0.0856
Cycle 2 plane: [0.31, 0.16, 0.94], residual 0.0884

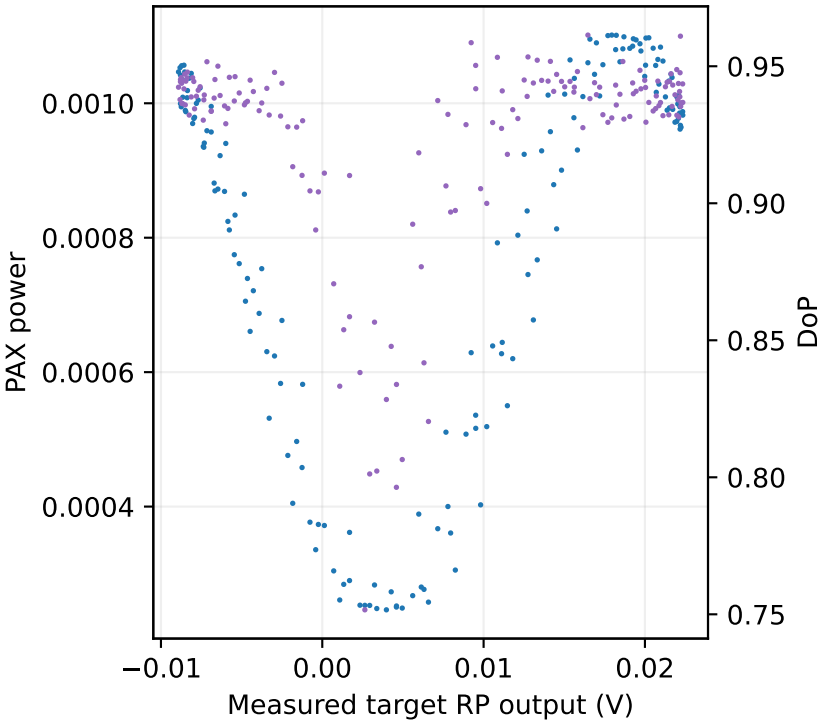
phi2 sine; fixed phi1 = 0.1549 RP V (segment 2)



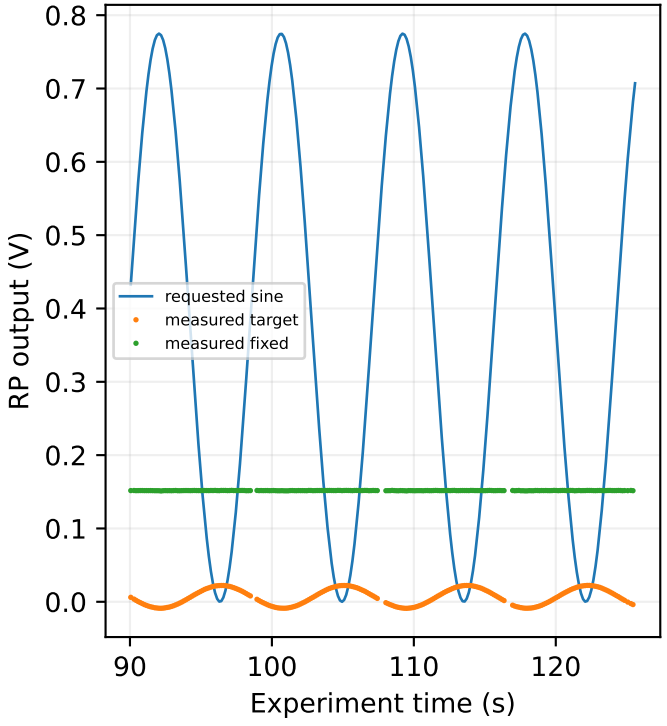
Stokes branches vs measured voltage



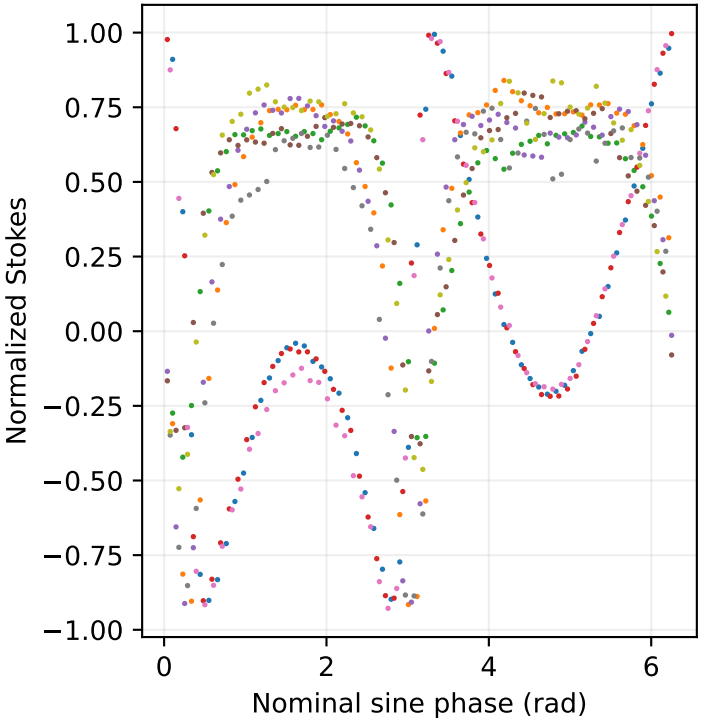
Power and DoP



Drive references and warm-up

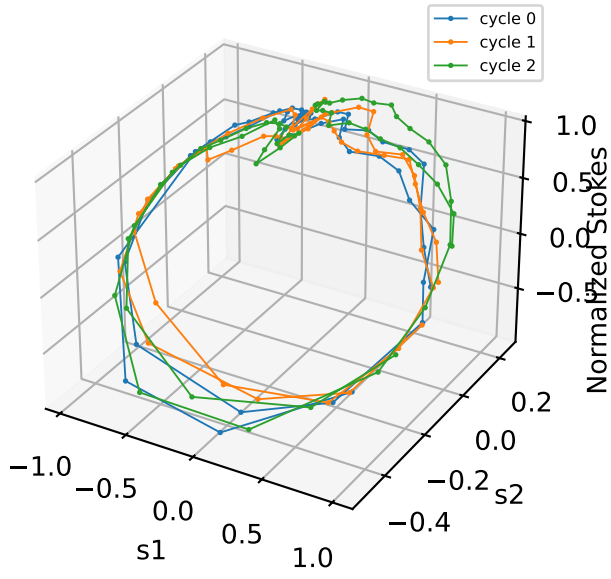


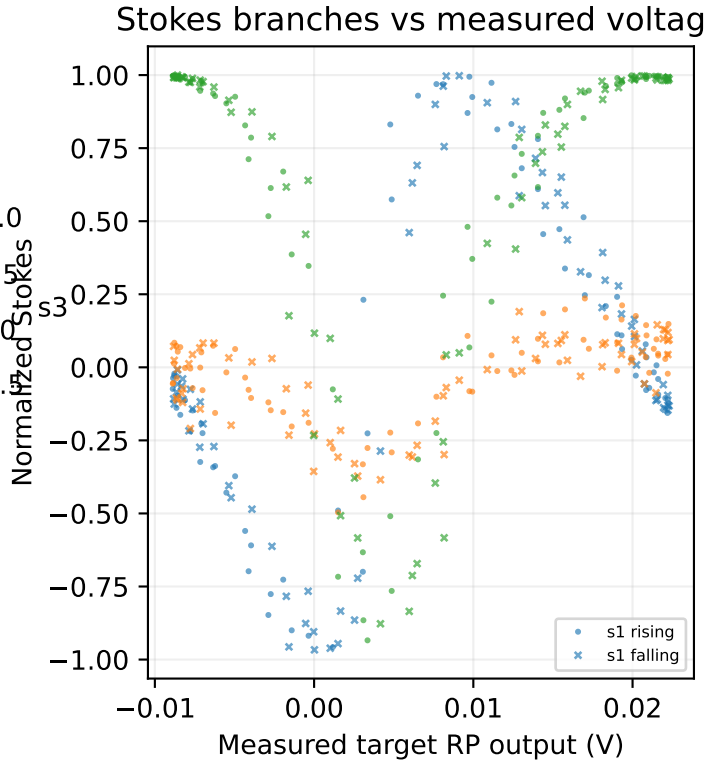
Cycle overlays

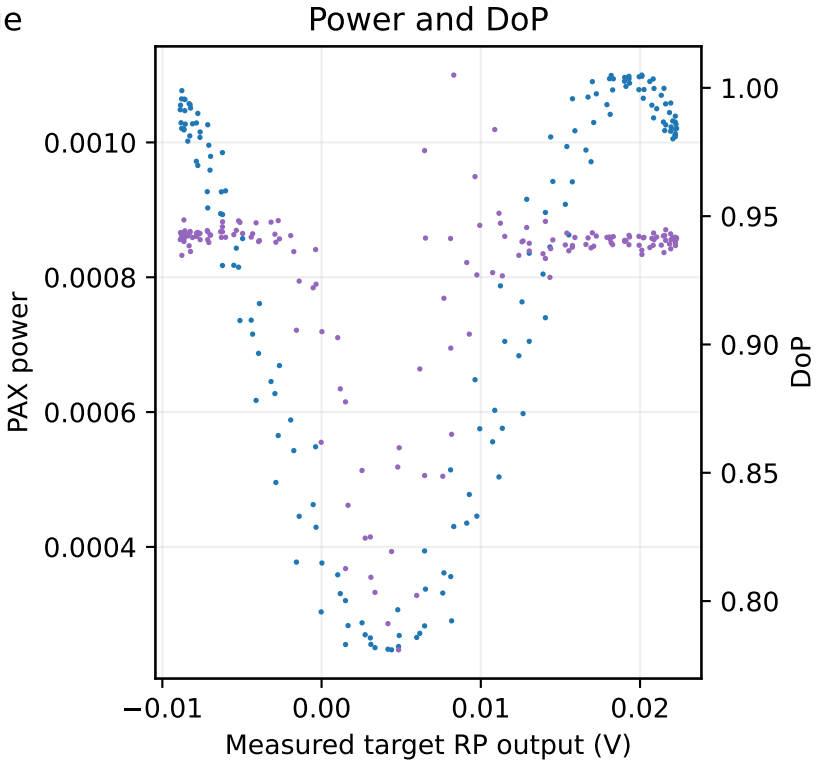


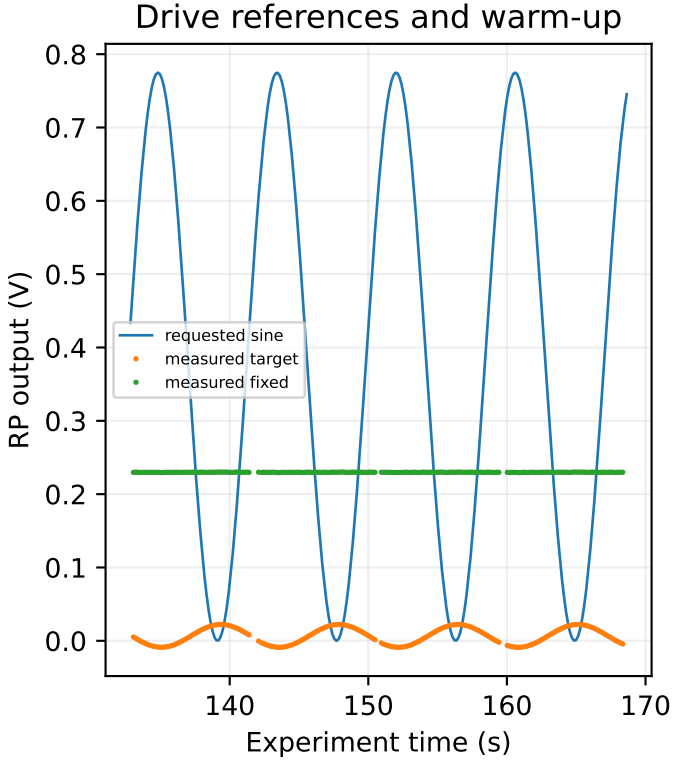
Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.248, -0.526, 0.814]
Plane residual RMS: 0.07228
Rising/falling angular RMS: 14.4°
Rising/falling power RMS: 6.92e-05 PAX units
Cycle angular RMS: 9.52°
Cycle power RMS: 3.17e-05 PAX units
Cycle DoP RMS: 0.0121
Tracking RMS: 0.4708 V
Reference coverage: 95.4%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.25, -0.48, 0.84], residual 0.0393
Cycle 1 plane: [0.24, -0.50, 0.83], residual 0.0589
Cycle 2 plane: [0.24, -0.60, 0.76], residual 0.0598

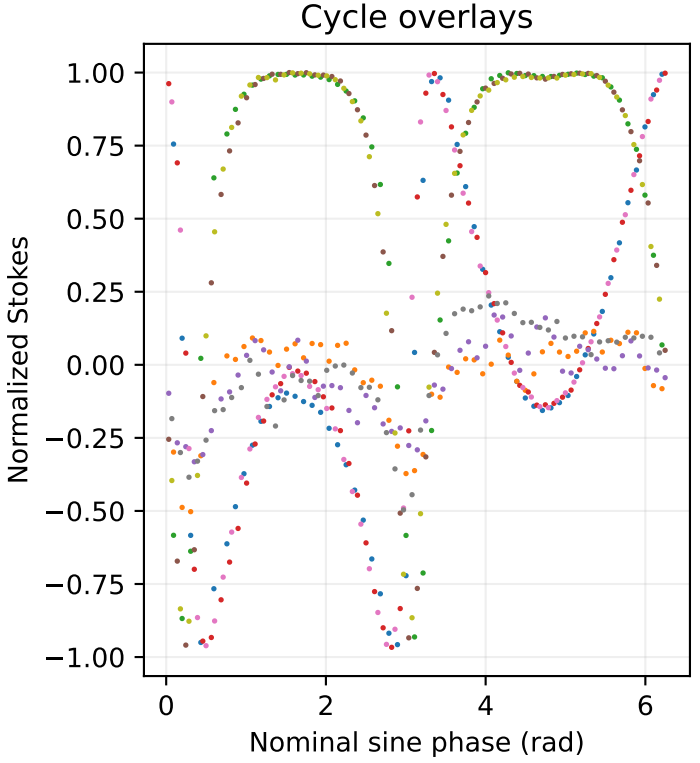
phi2 sine; fixed phi1 = 0.2324 RP V (segment 3)





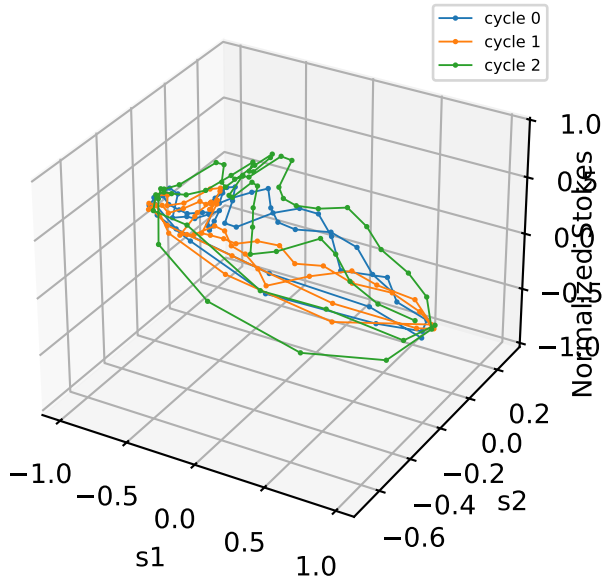


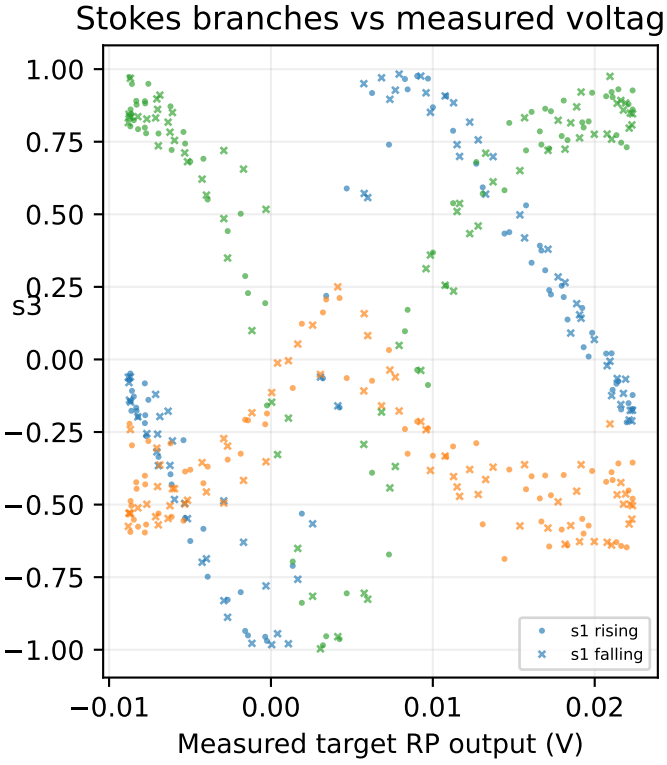


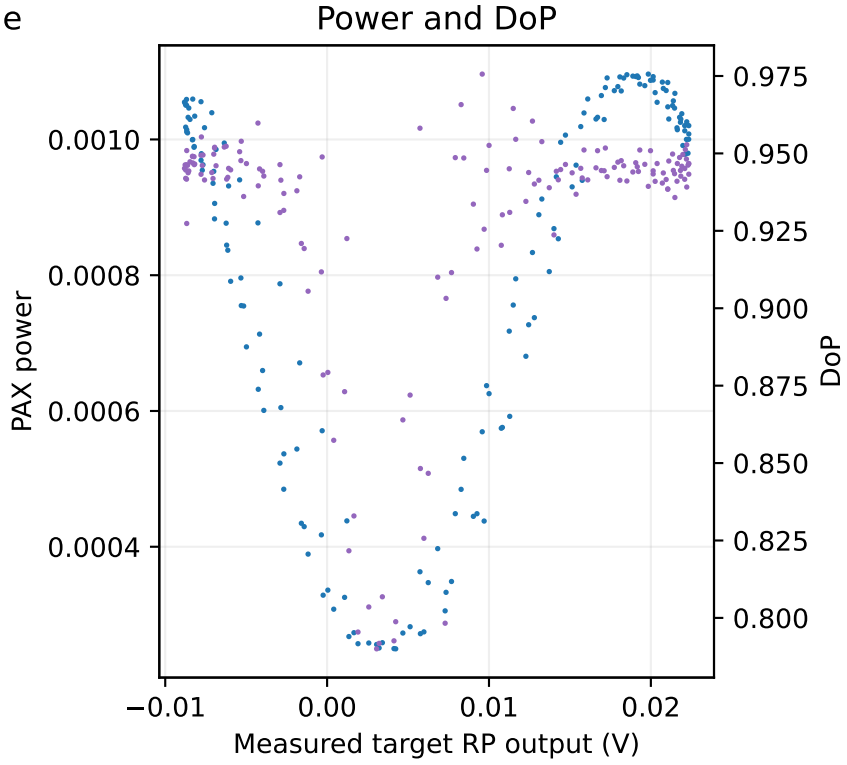


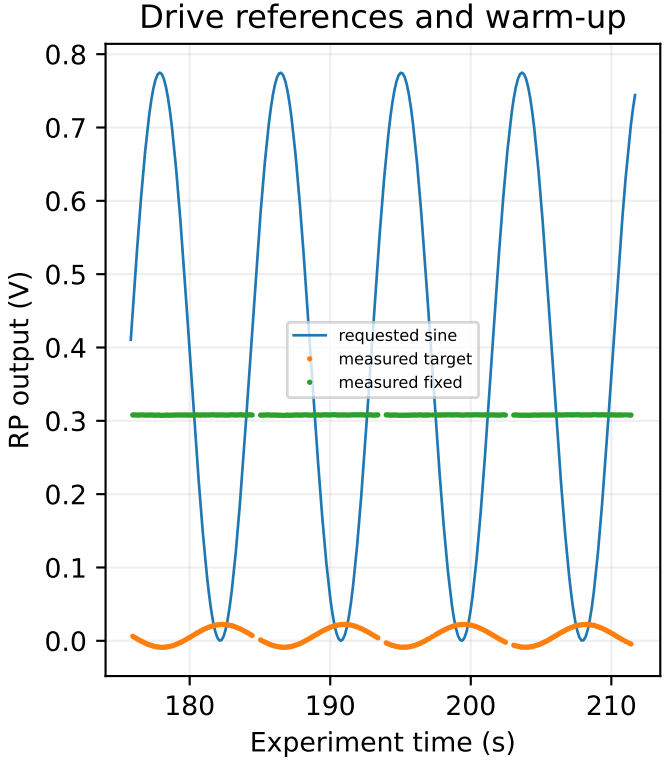
Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [-0.103, 0.969, -0.223]
Plane residual RMS: 0.07127
Rising/falling angular RMS: 19.3°
Rising/falling power RMS: 7.08e-05 PAX units
Cycle angular RMS: 10.4°
Cycle power RMS: 3.57e-05 PAX units
Cycle DoP RMS: 0.014
Tracking RMS: 0.4672 V
Reference coverage: 95.4%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [-0.07, 0.97, -0.25], residual 0.0471
Cycle 1 plane: [-0.09, 0.98, -0.18], residual 0.0649
Cycle 2 plane: [-0.15, 0.96, -0.24], residual 0.0803

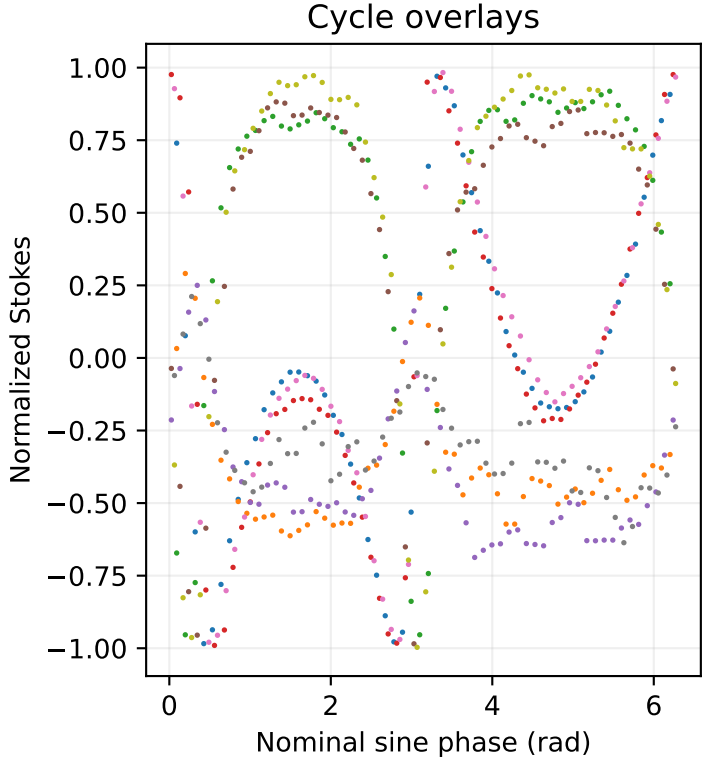
phi2 sine; fixed phi1 = 0.3099 RP V (segment 4)





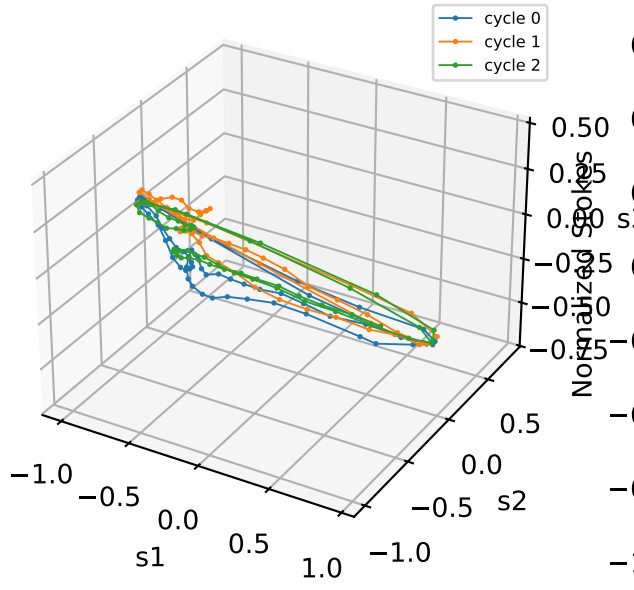


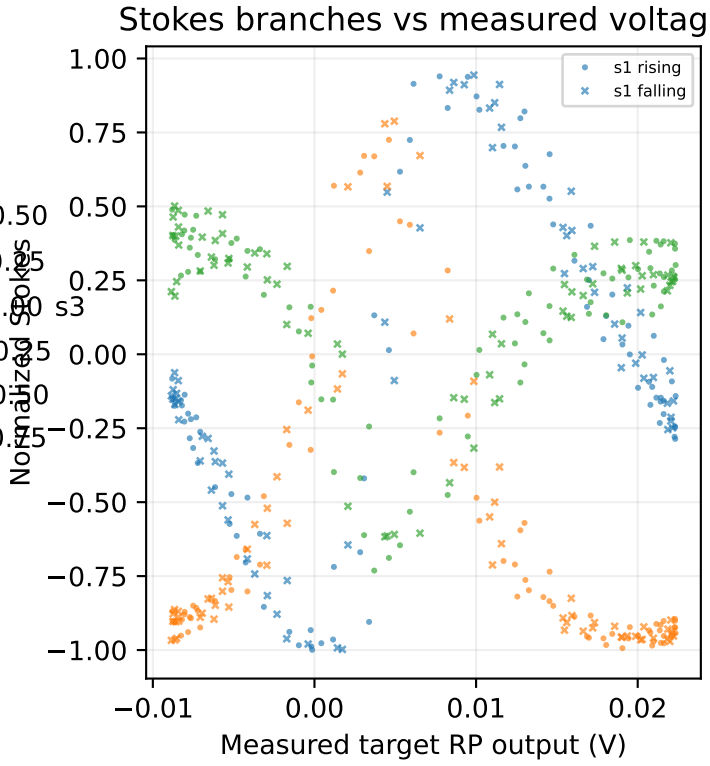


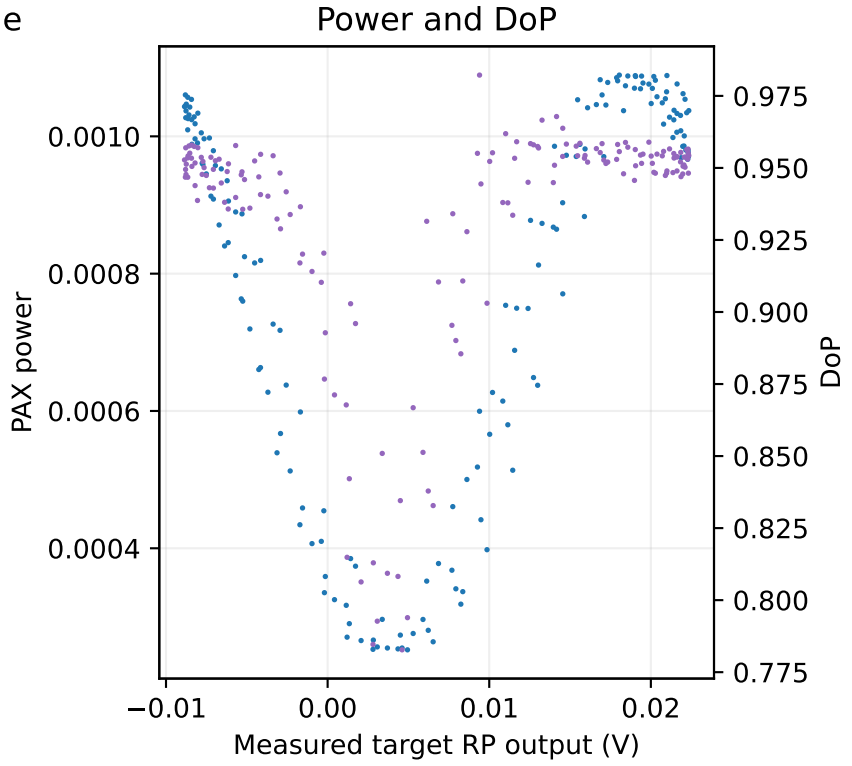


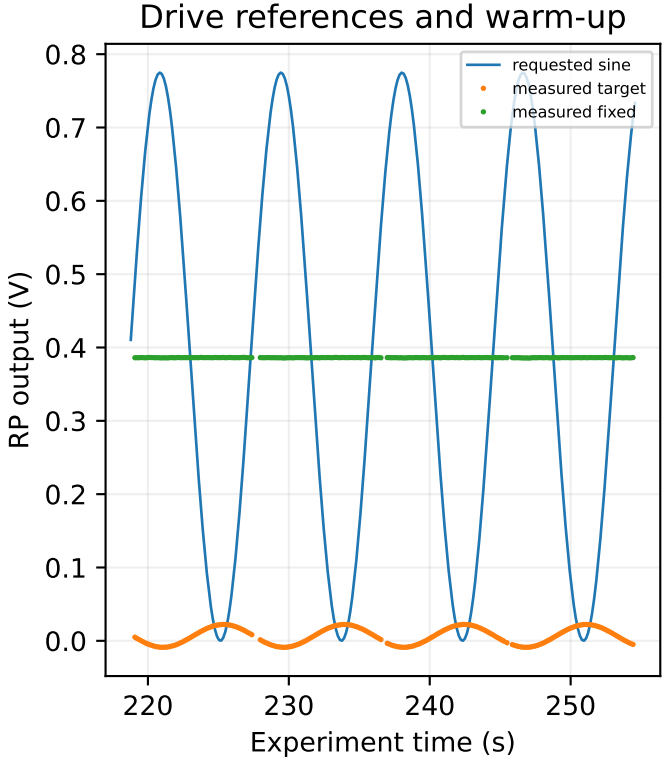
Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.038, 0.942, 0.334]
Plane residual RMS: 0.1026
Rising/falling angular RMS: 16.5°
Rising/falling power RMS: 6.73e-05 PAX units
Cycle angular RMS: 14.1°
Cycle power RMS: 4.75e-05 PAX units
Cycle DoP RMS: 0.025
Tracking RMS: 0.4676 V
Reference coverage: 94.9%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.02, 0.93, 0.37], residual 0.0626
Cycle 1 plane: [0.06, 0.91, 0.41], residual 0.0639
Cycle 2 plane: [0.04, 0.97, 0.23], residual 0.0956

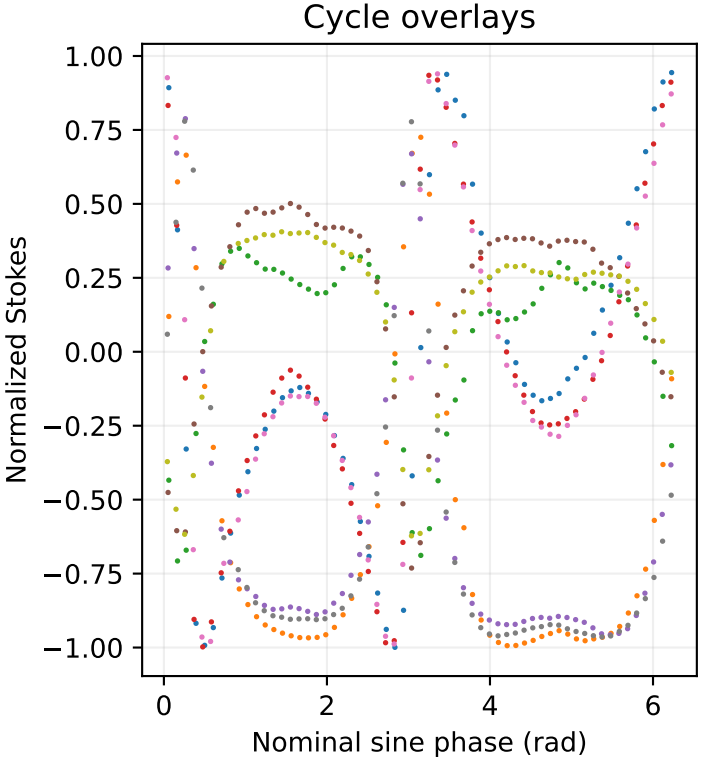
phi2 sine; fixed phi1 = 0.3875 RP V (segment 5)





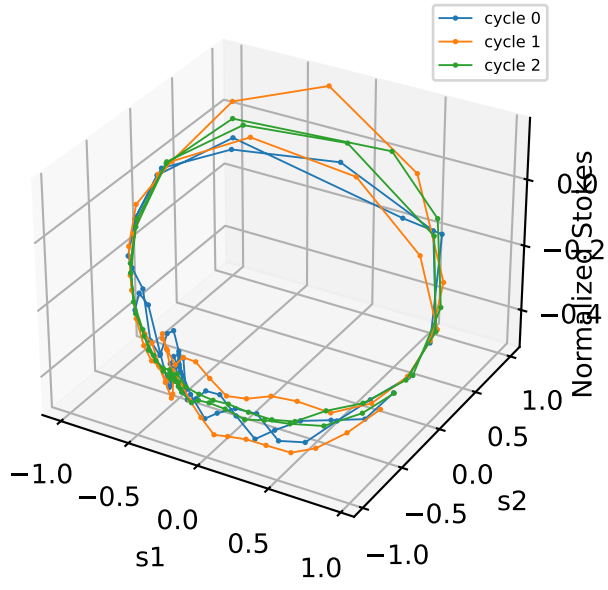




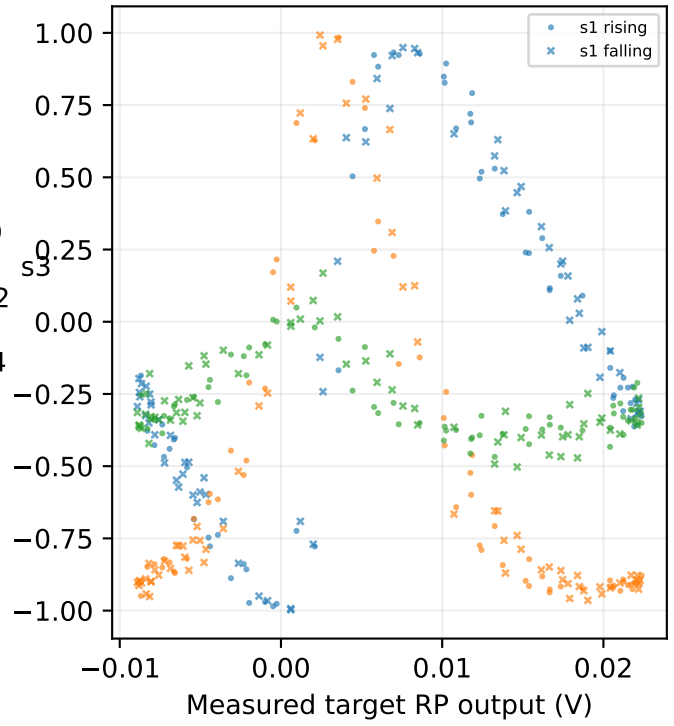


Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.164, 0.487, 0.858]
Plane residual RMS: 0.07169
Rising/falling angular RMS: 17.7°
Rising/falling power RMS: 5.73e-05 PAX units
Cycle angular RMS: 14.9°
Cycle power RMS: 7e-05 PAX units
Cycle DoP RMS: 0.0188
Tracking RMS: 0.47 V
Reference coverage: 96.6%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.16, 0.46, 0.87], residual 0.0598
Cycle 1 plane: [0.17, 0.53, 0.83], residual 0.0556
Cycle 2 plane: [0.15, 0.47, 0.87], residual 0.0427

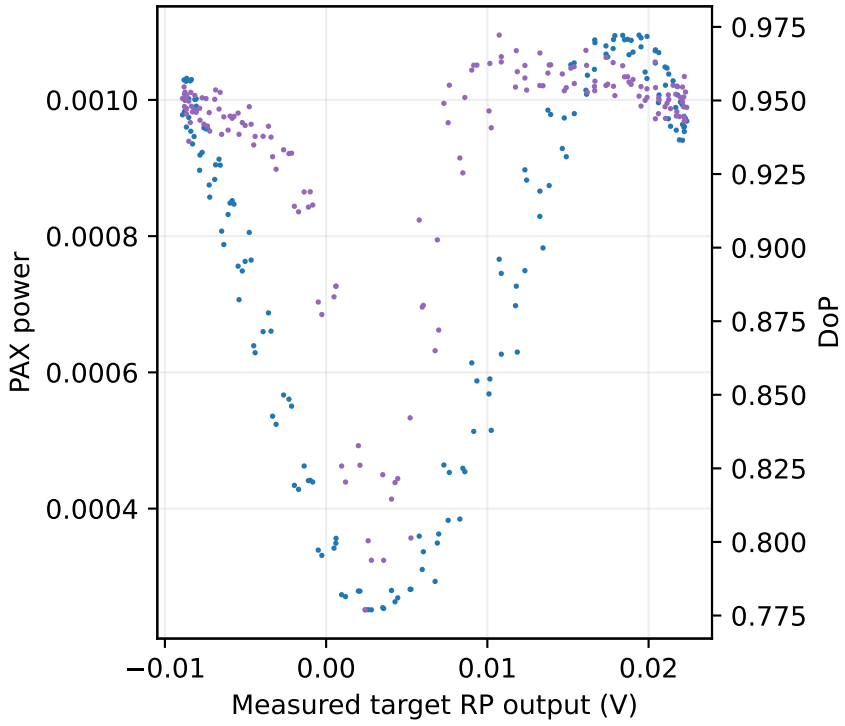
phi2 sine; fixed phi1 = 0.465 RP V (segment 6)



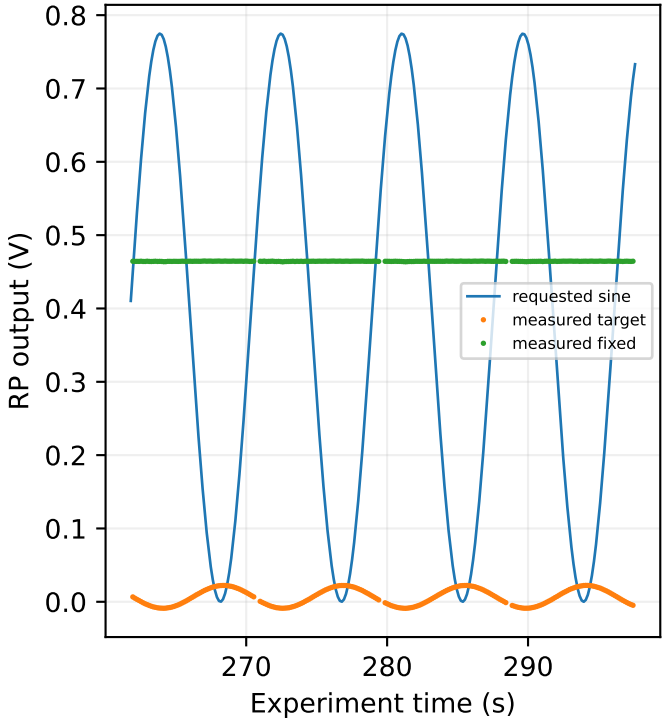
Stokes branches vs measured voltage



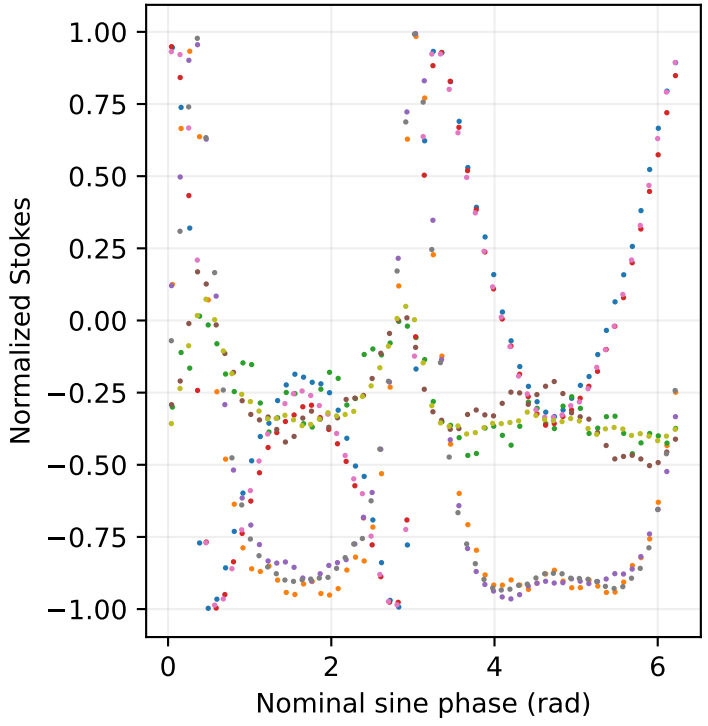
Power and DoP



Drive references and warm-up

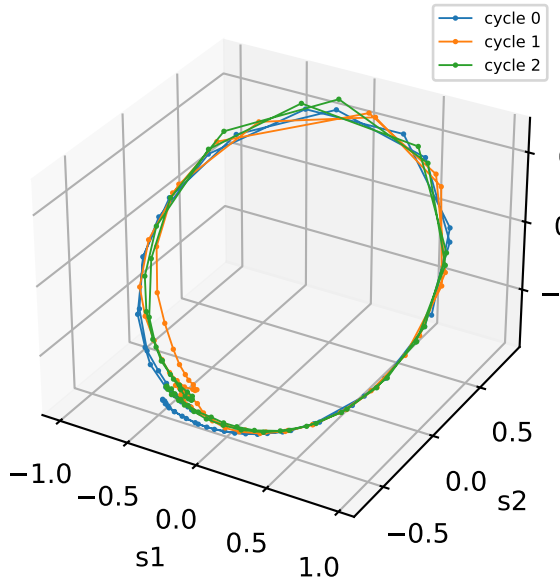


Cycle overlays

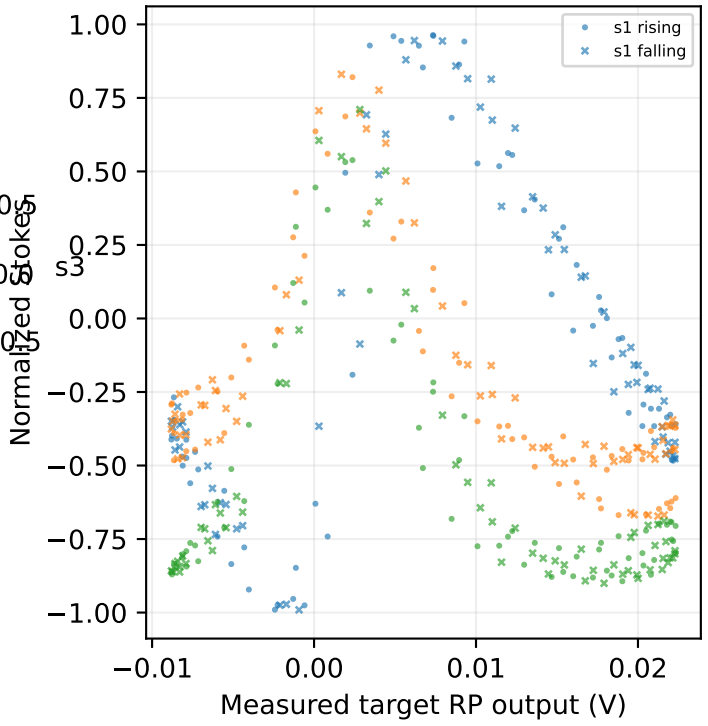


Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.145, -0.186, 0.972]
Plane residual RMS: 0.05477
Rising/falling angular RMS: 10.5°
Rising/falling power RMS: 6.11e-05 PAX units
Cycle angular RMS: 9.86°
Cycle power RMS: 4.32e-05 PAX units
Cycle DoP RMS: 0.0107
Tracking RMS: 0.4708 V
Reference coverage: 96.6%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.16, -0.16, 0.97], residual 0.0563
Cycle 1 plane: [0.15, -0.19, 0.97], residual 0.0676
Cycle 2 plane: [0.13, -0.20, 0.97], residual 0.0216

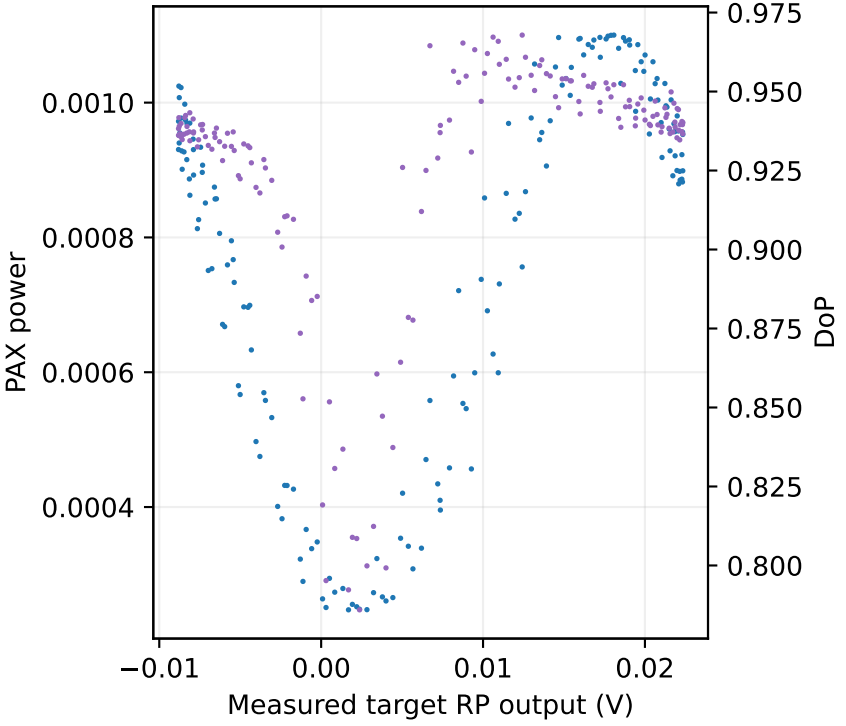
phi2 sine; fixed phi1 = 0.5425 RP V (segment 7)



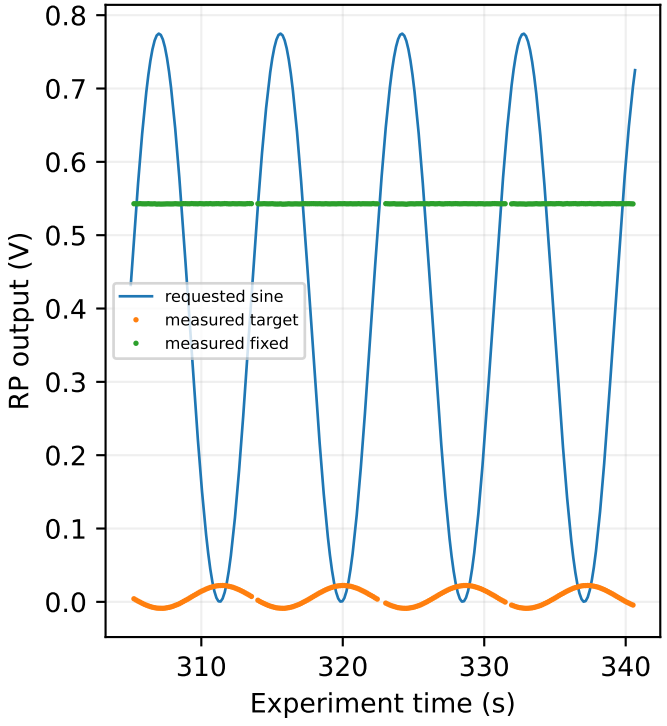
Stokes branches vs measured voltage



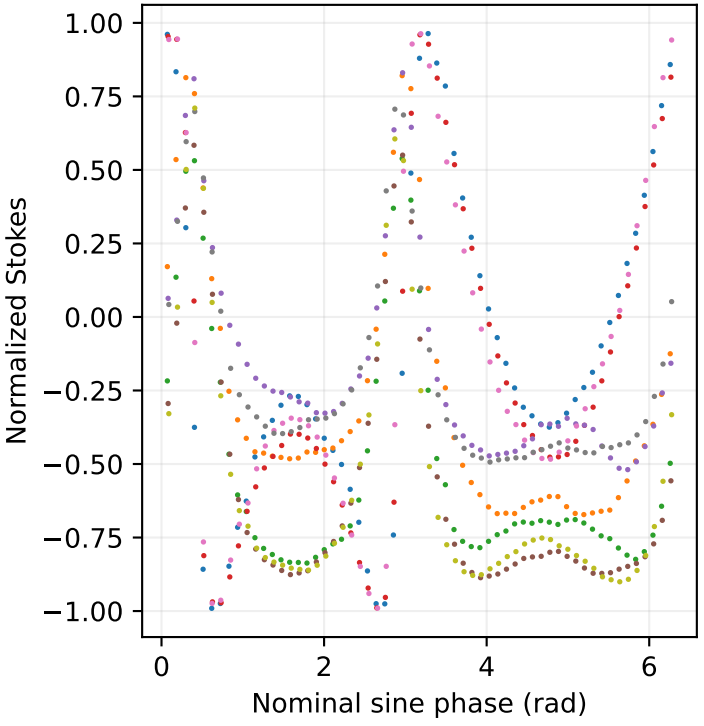
Power and DoP



Drive references and warm-up

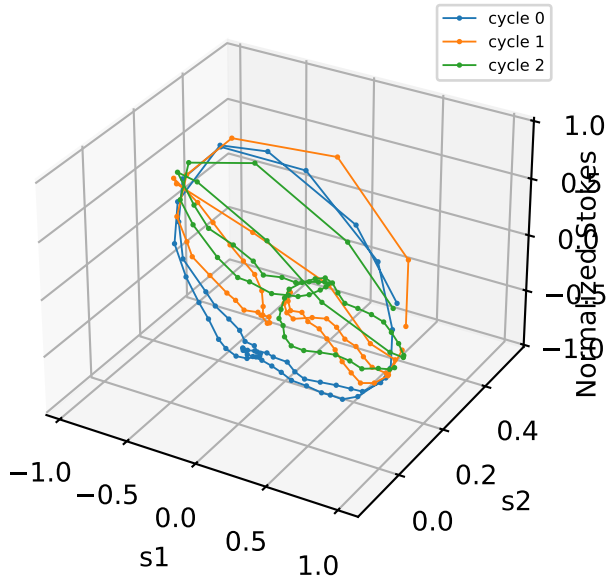


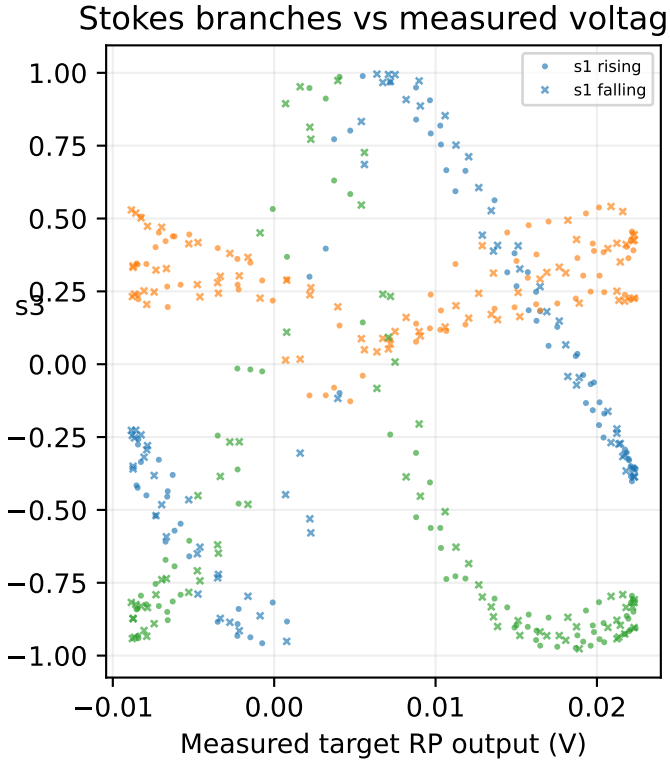
Cycle overlays

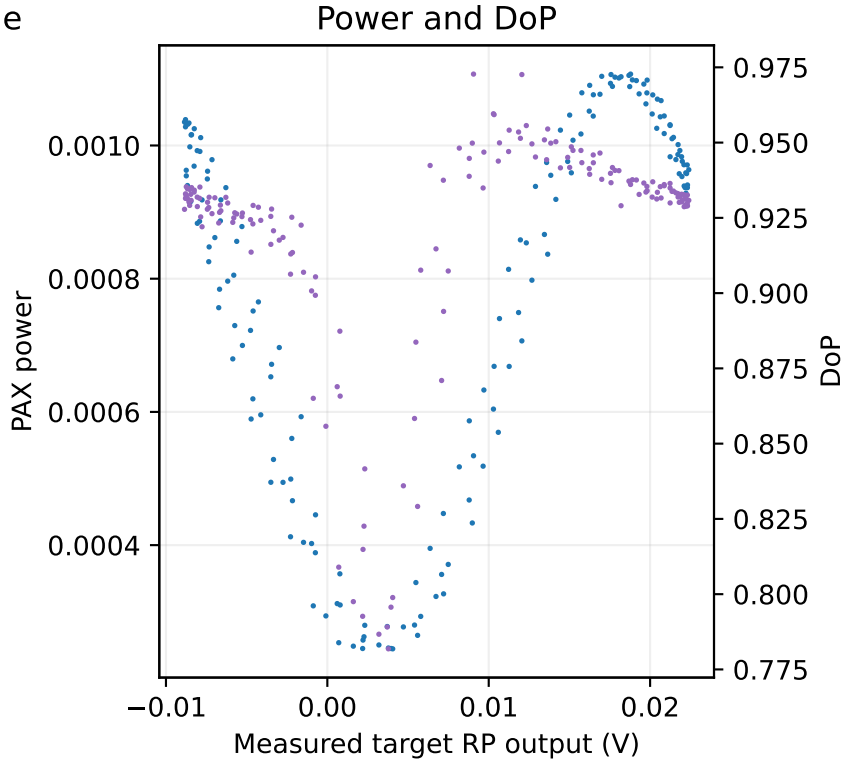


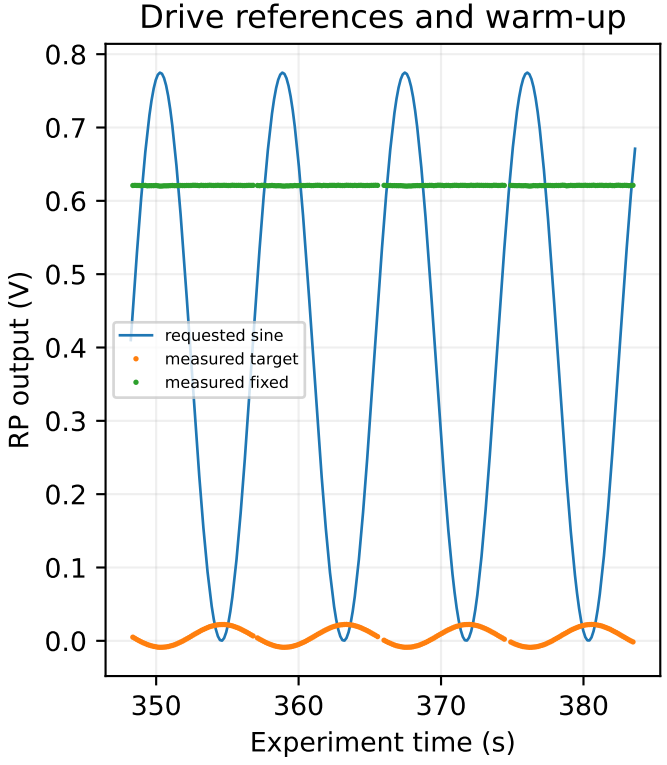
Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [-0.029, 0.747, -0.664]
Plane residual RMS: 0.08903
Rising/falling angular RMS: 15.3°
Rising/falling power RMS: 8.1e-05 PAX units
Cycle angular RMS: 13.4°
Cycle power RMS: 5.98e-05 PAX units
Cycle DoP RMS: 0.0108
Tracking RMS: 0.4705 V
Reference coverage: 96.0%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.06, -0.68, 0.73], residual 0.073
Cycle 1 plane: [-0.00, 0.77, -0.63], residual 0.062
Cycle 2 plane: [-0.04, 0.79, -0.61], residual 0.0369

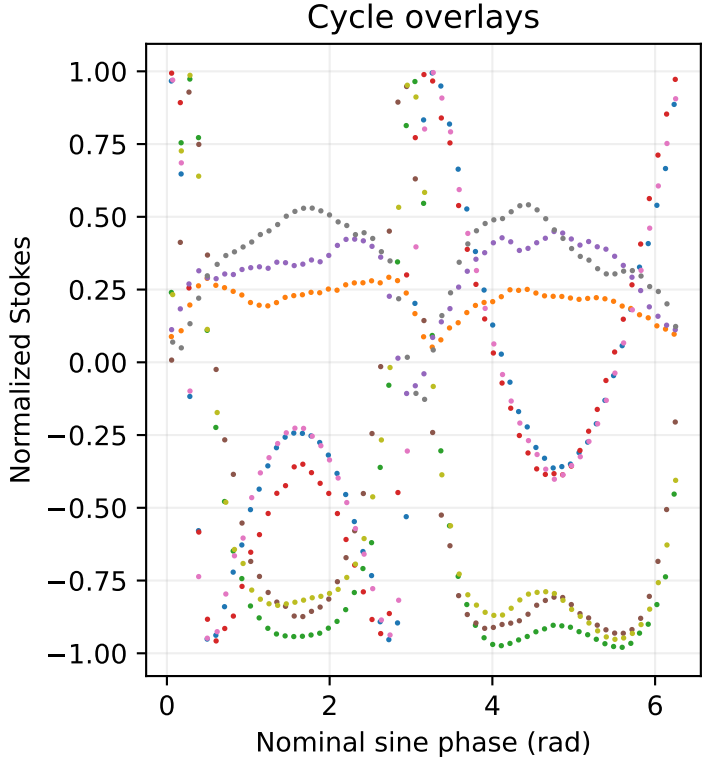
phi2 sine; fixed phi1 = 0.62 RP V (segment 8)





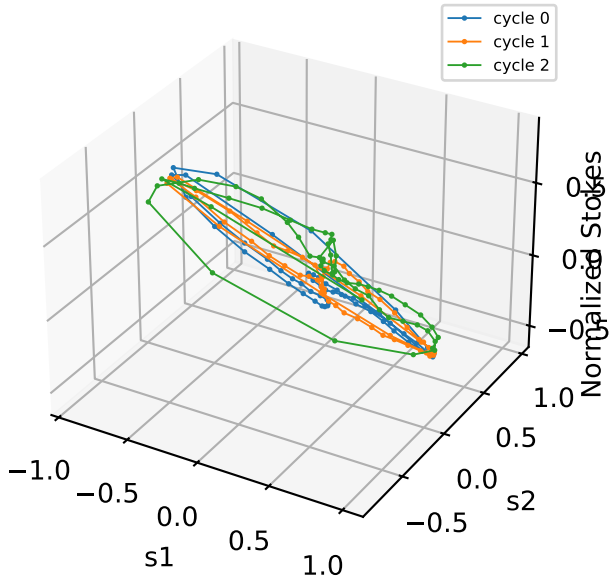


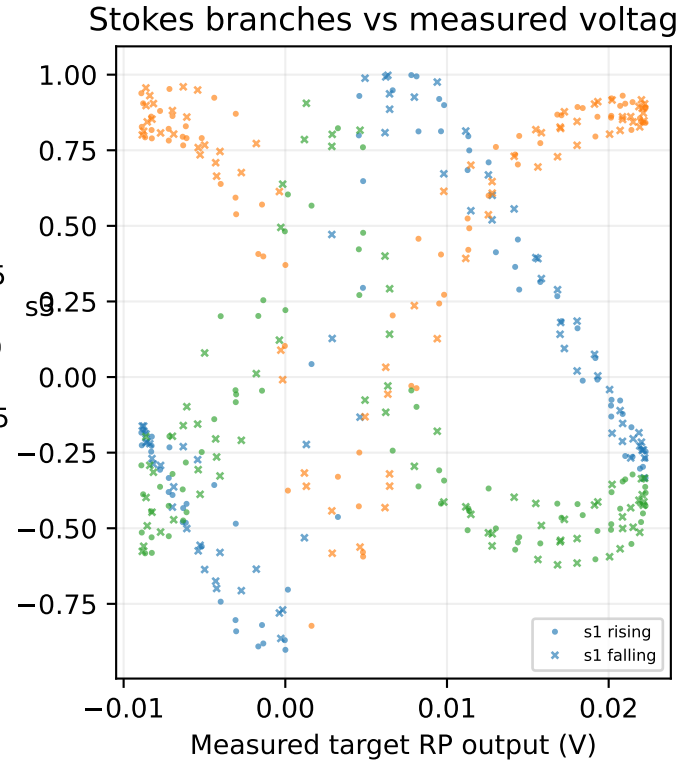


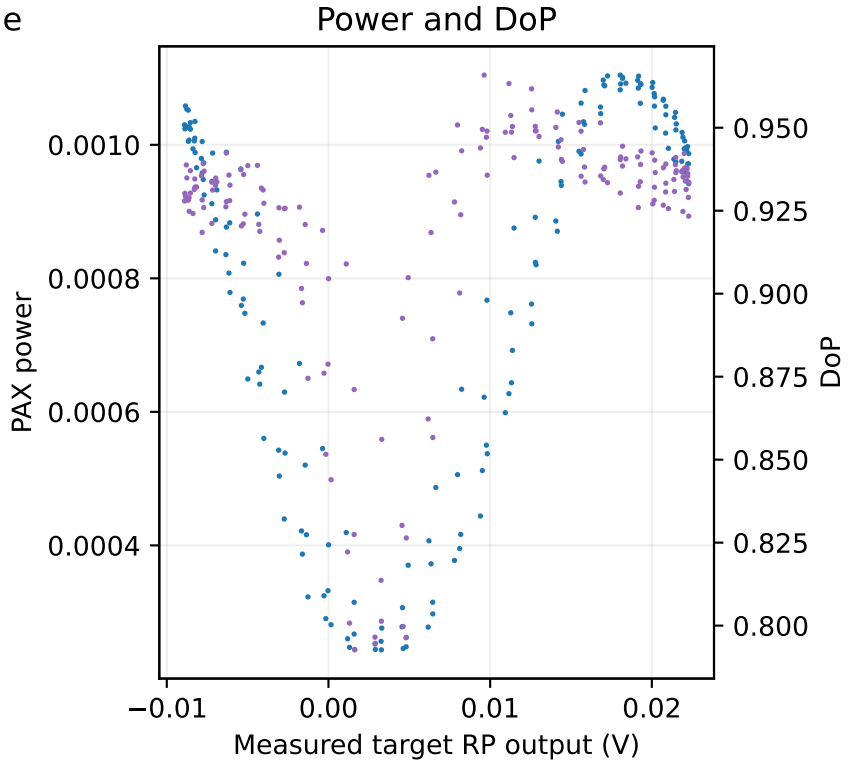


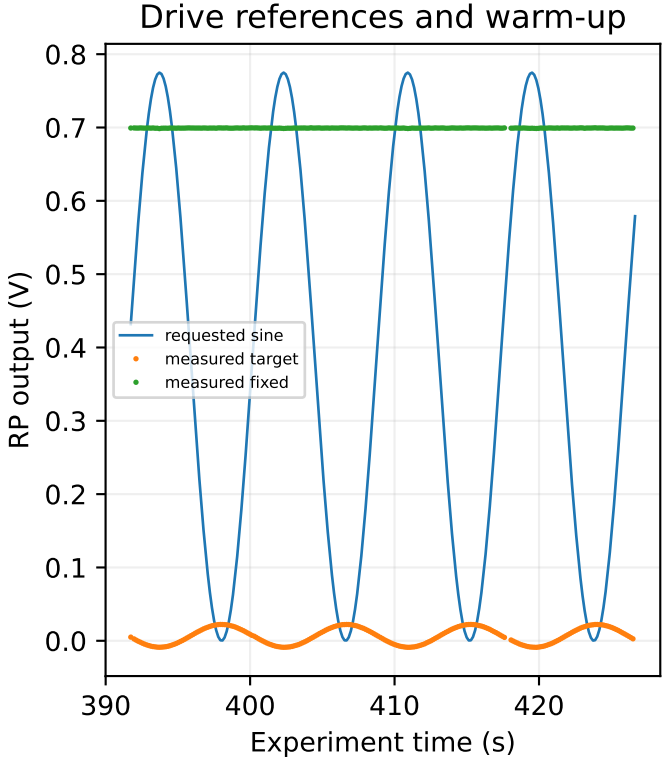
Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.128, 0.985, 0.118]
Plane residual RMS: 0.09613
Rising/falling angular RMS: 29.9°
Rising/falling power RMS: 8.51e-05 PAX units
Cycle angular RMS: 14.8°
Cycle power RMS: 5.9e-05 PAX units
Cycle DoP RMS: 0.0121
Tracking RMS: 0.4736 V
Reference coverage: 97.2%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.10, 0.99, 0.01], residual 0.0195
Cycle 1 plane: [0.13, 0.98, 0.15], residual 0.0646
Cycle 2 plane: [0.14, 0.97, 0.21], residual 0.07

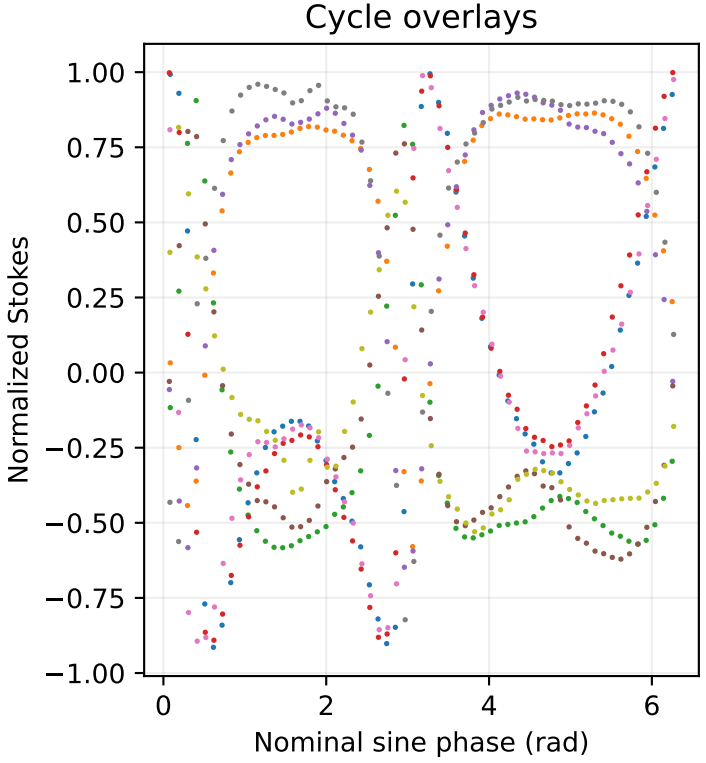
phi2 sine; fixed phi1 = 0.6974 RP V (segment 9)





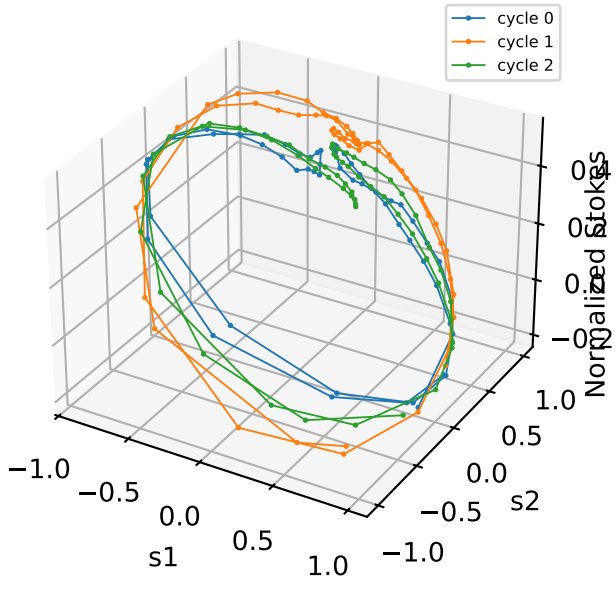




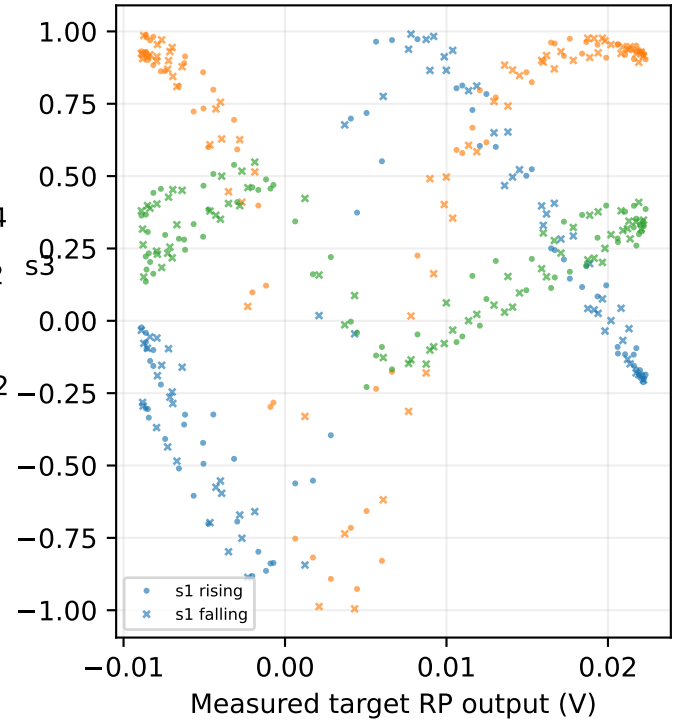


Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.264, 0.638, 0.724]
Plane residual RMS: 0.09364
Rising/falling angular RMS: 24.7°
Rising/falling power RMS: 8.39e-05 PAX units
Cycle angular RMS: 18.3°
Cycle power RMS: 5.34e-05 PAX units
Cycle DoP RMS: 0.0196
Tracking RMS: 0.4749 V
Reference coverage: 98.9%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.22, 0.71, 0.66], residual 0.0471
Cycle 1 plane: [0.26, 0.65, 0.72], residual 0.0616
Cycle 2 plane: [0.30, 0.53, 0.79], residual 0.0707

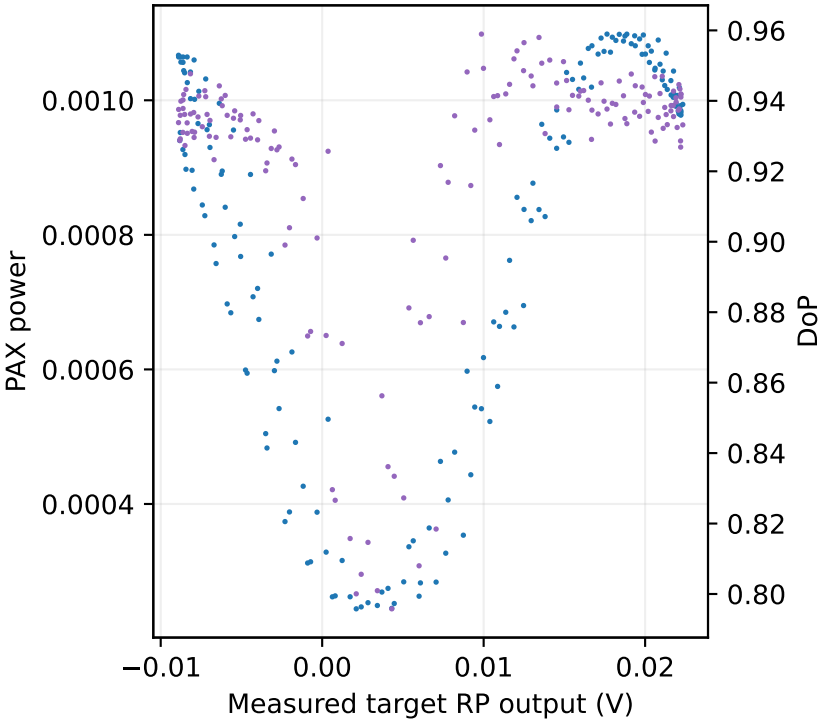
phi2 sine; fixed phi1 = 0.7749 RP V (segment 10)



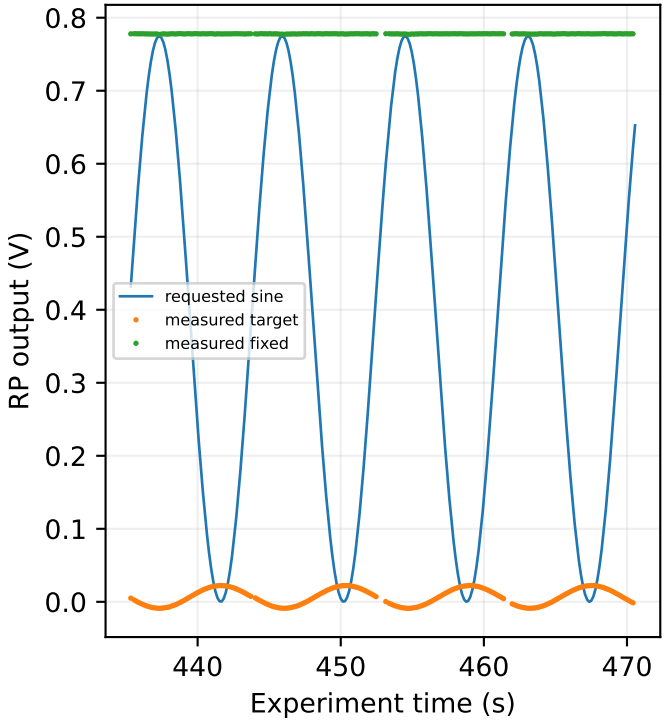
Stokes branches vs measured voltage



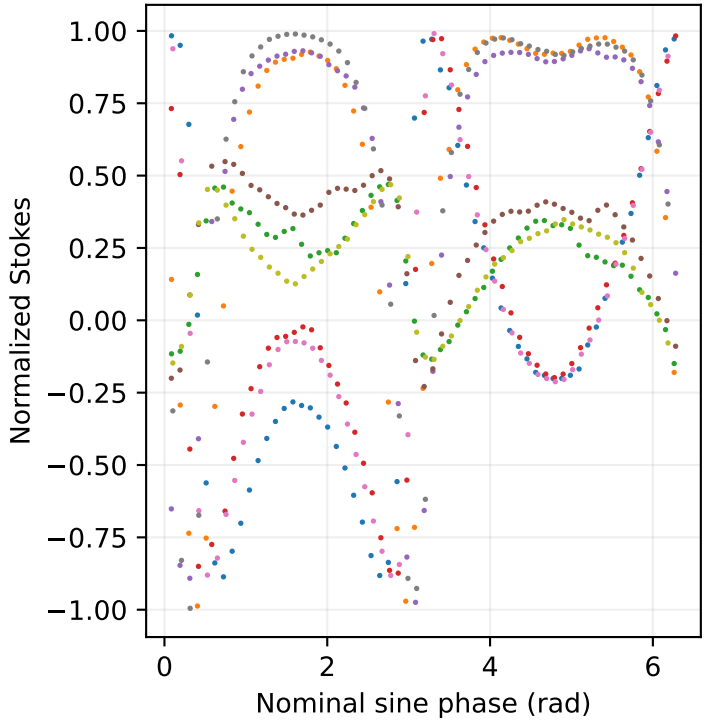
Power and DoP



Drive references and warm-up



Cycle overlays

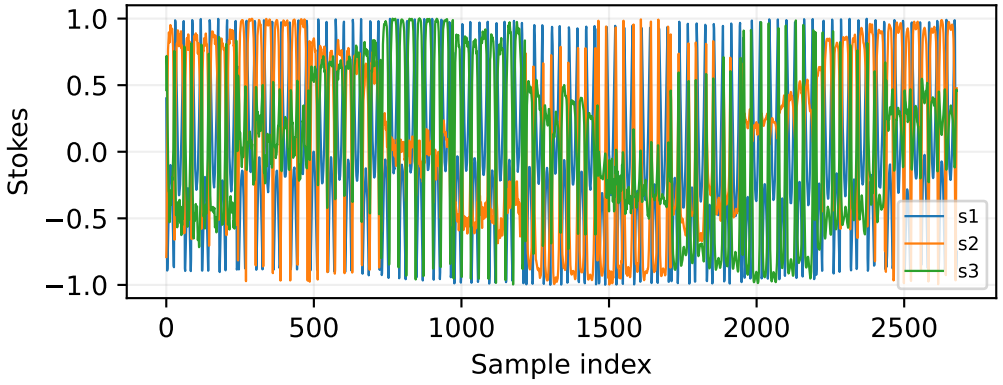


Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.283, -0.096, 0.954]
Plane residual RMS: 0.08053
Rising/falling angular RMS: 29.3°
Rising/falling power RMS: 8.78e-05 PAX units
Cycle angular RMS: 21.4°
Cycle power RMS: 0.00011 PAX units
Cycle DoP RMS: 0.0259
Tracking RMS: 0.4725 V
Reference coverage: 96.0%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.28, -0.01, 0.96], residual 0.0429
Cycle 1 plane: [0.28, -0.22, 0.93], residual 0.0287
Cycle 2 plane: [0.28, -0.06, 0.96], residual 0.0502

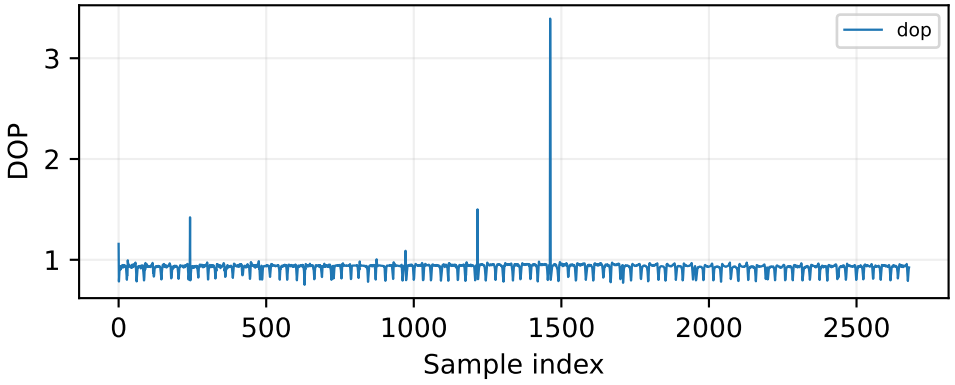
Recorded telemetry

Acquisition order preserves stage changes; no timing alignment or transfer gain is inferred.

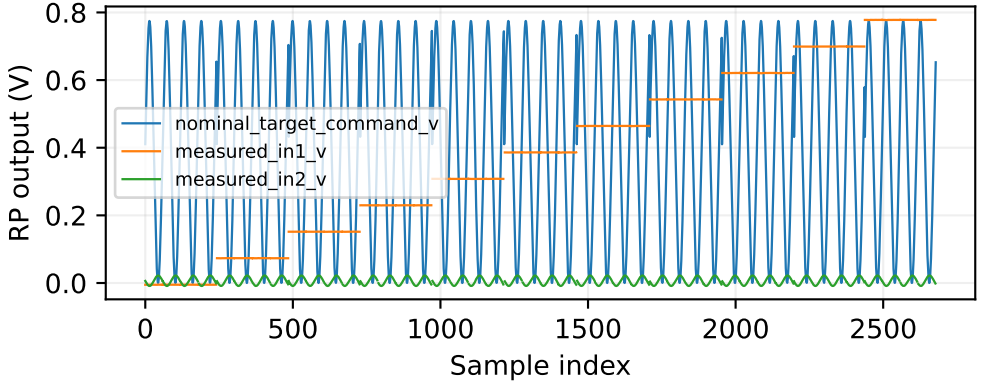
Normalized Stokes



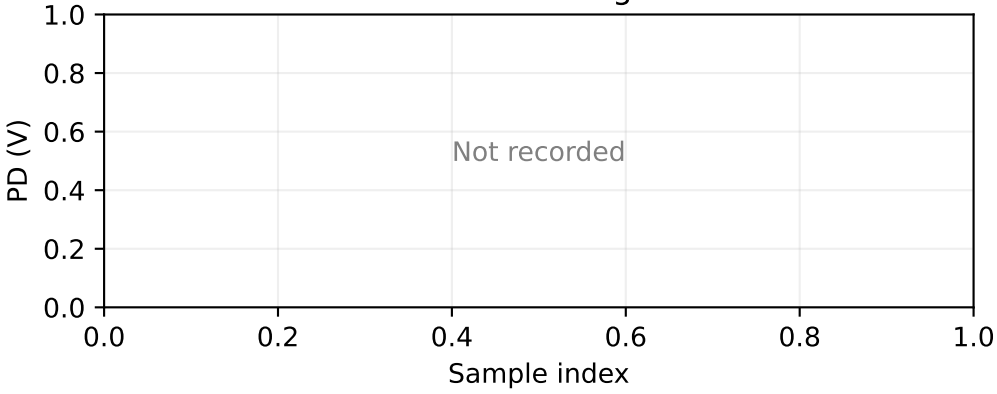
Raw degree of polarization



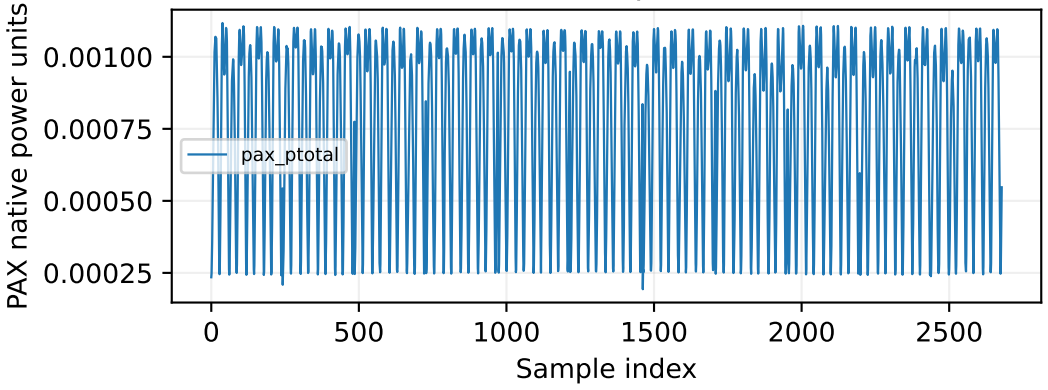
Actuator command / reference history



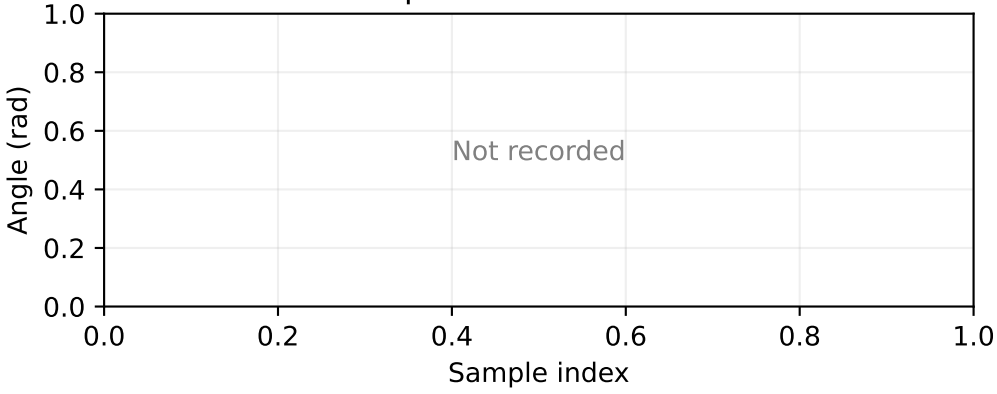
Photodiode signal



PAX total power



Sphere coordinates



Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
label	"quick-test-phi2"
comment	"dry run 2 except with phi2 now"
actuator	"phi2"
report	"pdf"
pax_rotation_velocity_hz	60.0
pax_measurement_mode	5
pax_wavelength_nm	830.0
phi1_v_lambda	13.078125
phi2_v_lambda	13.078125
phi1_actuator_volts_per_rp_volt	16.875
phi2_actuator_volts_per_rp_volt	16.875
rp_output_min_voltage	0.0
rp_output_max_voltage	1.0
cross_sweep_sine_frequency_hz	0.1
cross_sweep_sine_center_voltage	0.3875
cross_sweep_sine_amplitude_voltage	0.3875
cross_sweep_warmup_cycles	1
cross_sweep_recorded_cycles	3
cross_sweep_phi1_bias_voltages	[0.0, 0.0775, 0.155, 0.2325, 0.31, 0.3875, 0.465, 0.5425, 0.62, 0
cross_sweep_phi2_bias_voltages	[0.0, 0.0775, 0.155, 0.2325, 0.31, 0.3875, 0.465, 0.5425, 0.62, 0
cross_sweep_settle_s	1.0
cross_sweep_bias_ramp_s	1.0
cross_sweep_bias_ramp_updates_per_s	50.0
cross_sweep_reference_sample_rate_hz	400.0

Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
cross_sweep_scope_block_s	8.0
cross_sweep_min_pax_samples_per_cycle	20.0